AI Methods for Turbulence and Early-Universe Signals

A KAAI Workshop · Carnegie Mellon University

November 2 – 6 · McWilliams Center for Cosmology & Astrophysics, CMU

SIMONS FOUNDATION

McWilliams Center for Cosmology & Astrophysics

WHAT CAN WE LEARN FROM PULSAR TIMING ARRAYS WHEN AI IS ON BOARD?

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Carnegie Mellon University, USA and Ilia State University, Georgia

and Abastumani Astrophysical Observatory, Georgia

November 2, 2026

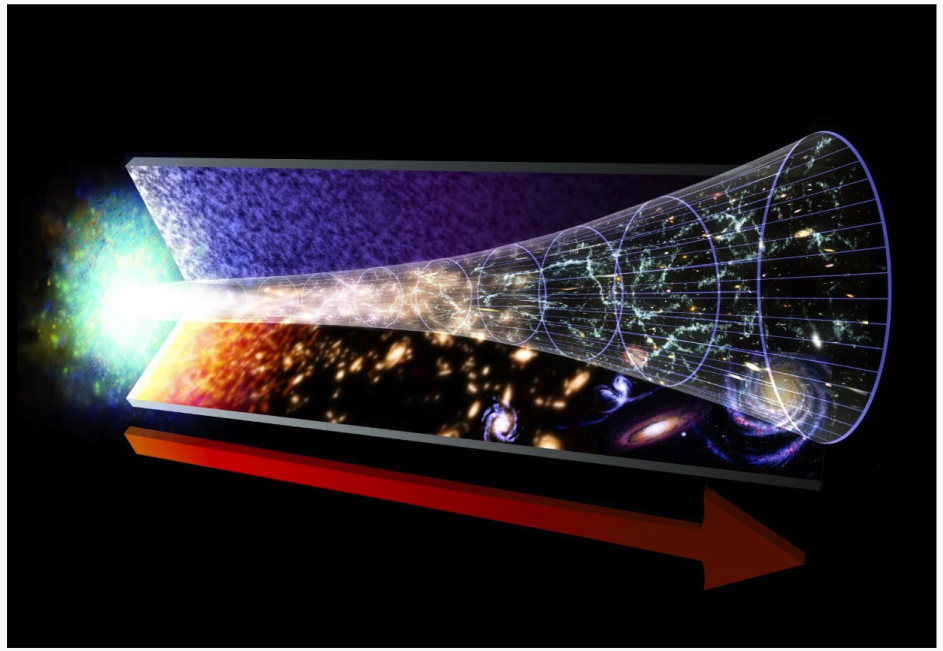
OUTLINE

- **Introduction:** Why Gravitational Waves - Why New Physics?
- **Fundamental Physics**: From the First Seconds and Even Earlier
- **Pulsar Timing Arrays:** Stochastic Gravitational Waves Background and Early Universe Source
- **Anisotropic Phases in the Early Universe:** Bubbles, Sound Waves, Turbulence
- **Going Beyond GR**: Massive Gravity
- **Why PTA & AI are important:** What Can PTA Teach Us and What Can AI Teach Us
- **Take Home**

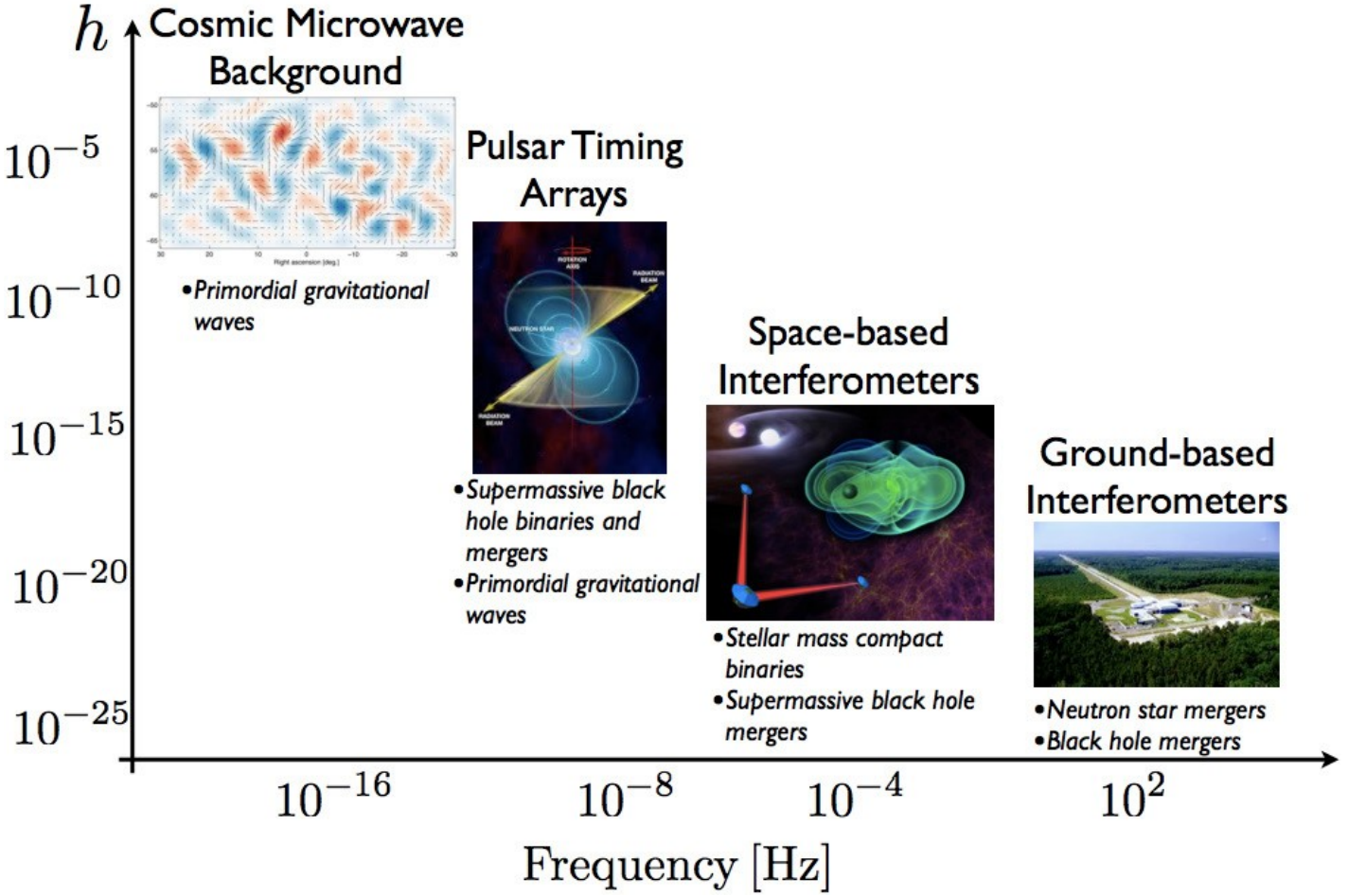
COLLABORATION

- **Seniors and Juniors:** Axel Brandenburg, Grigol Gogoberidze, Arthur Kosowsky, Andrew Long, Sayan Mandal, Barnabas Poczos, Alberto Roper Pol
- **Students**: Murman Bosca, Chris Choi, Emma Clarke, Murman Gurgenidze, Ved Kenjale, Nakul Shenoy, Jonathan Stepp

WHY GRAVITATIONAL WAVES?

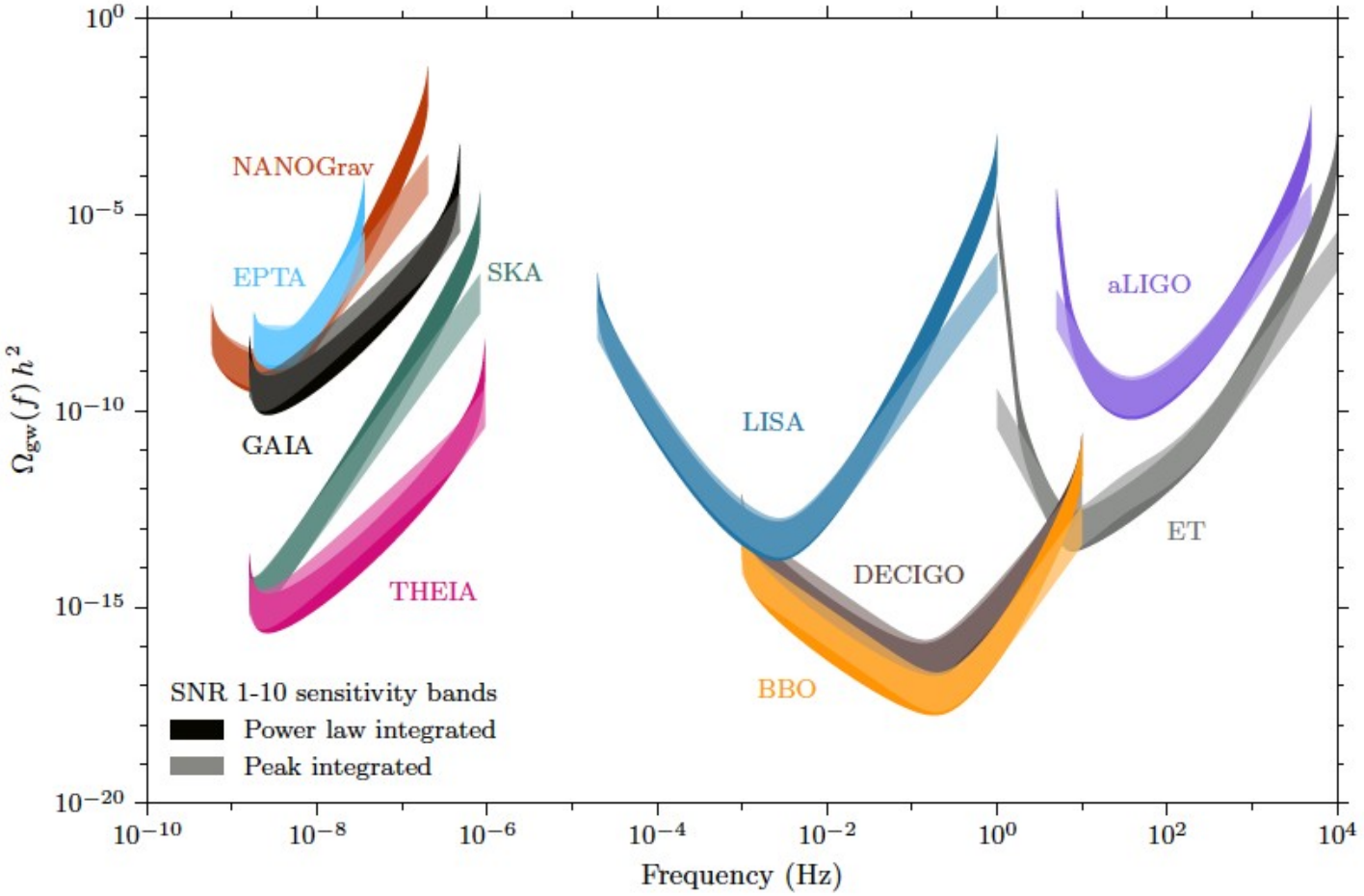


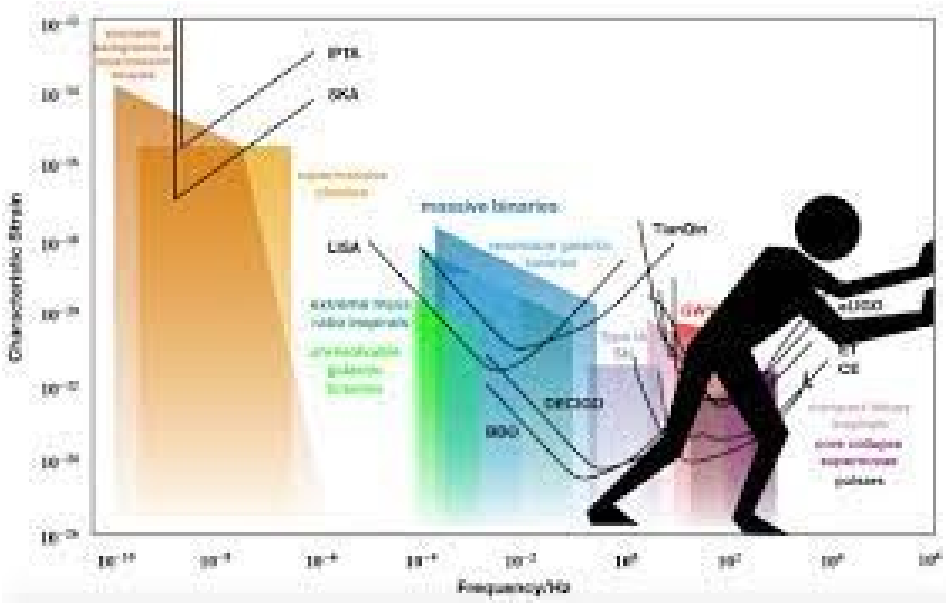
Artist's impression of the "Arrow of Time". Source: NASA/GSFC



WHAT IS NEXT?

Garcia Bellido et al. 2020

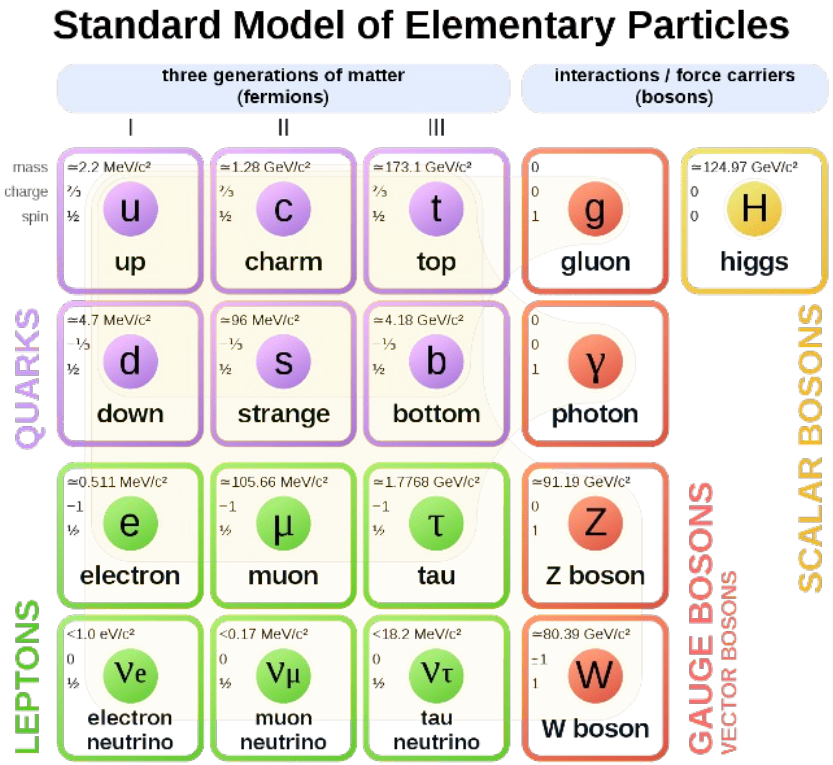


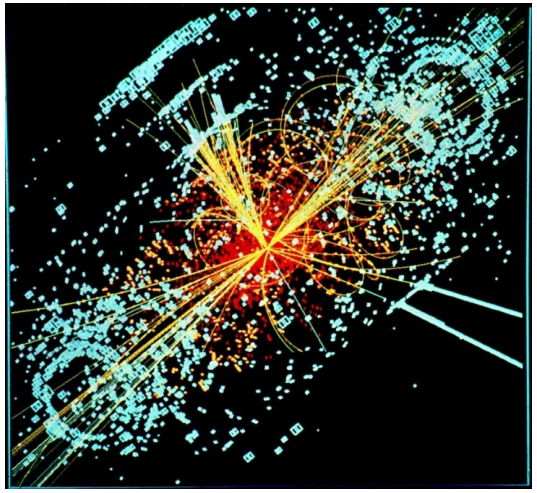


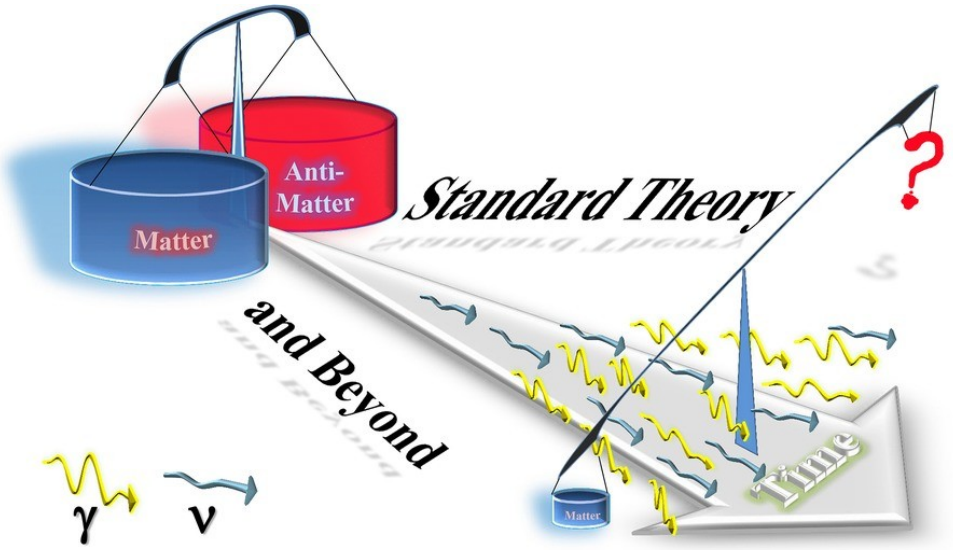
Credit CERN

Connection with High Energy Particle Physics – the best laboratory to test the energy scales EVEN near the Planck scale

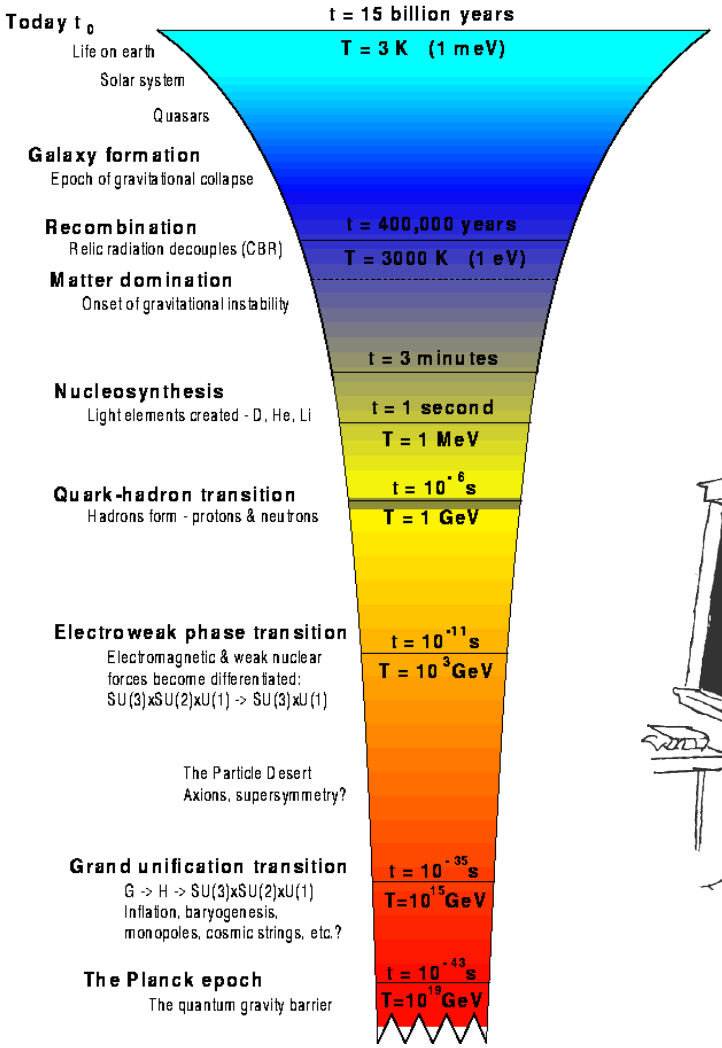
WHY NEW PHYSICS (PARTICLES AND FIELDS)?

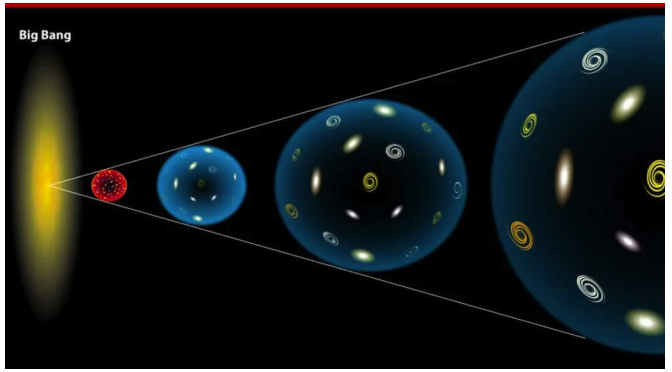


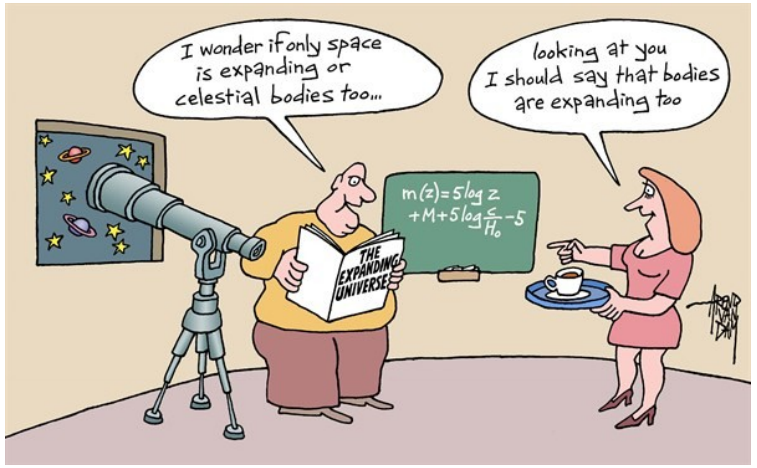


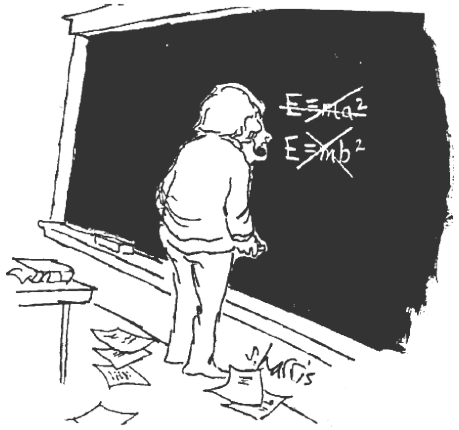


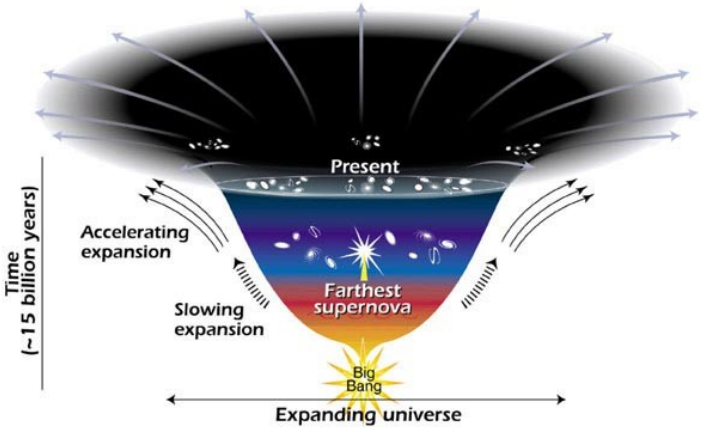
WHY NEW PHYSICS (EARLY AND LATE UNIVERSE)?





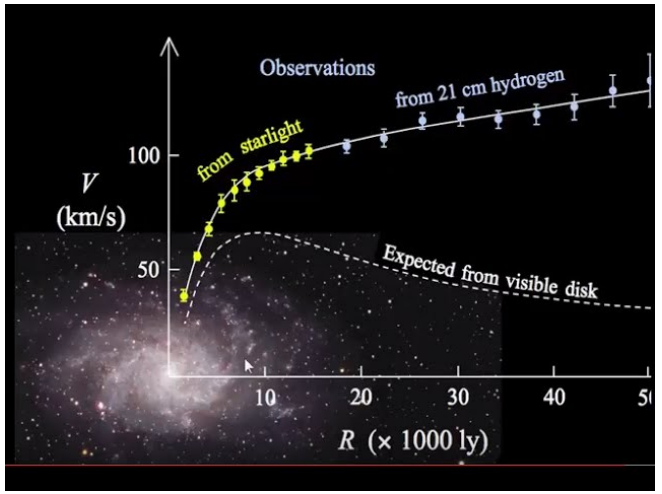


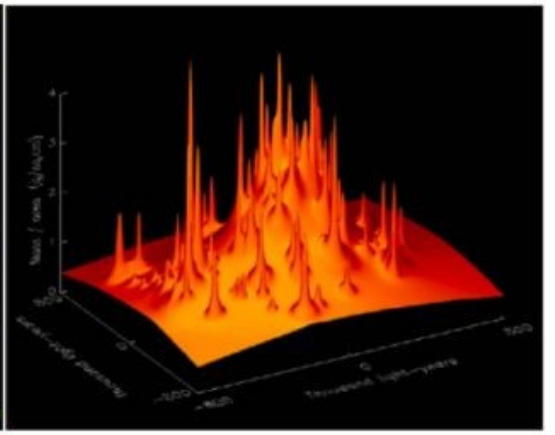


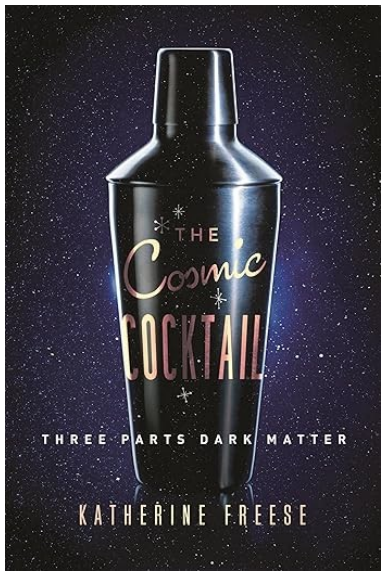


WHY NEW PHYSICS (IN THE DARK) ?

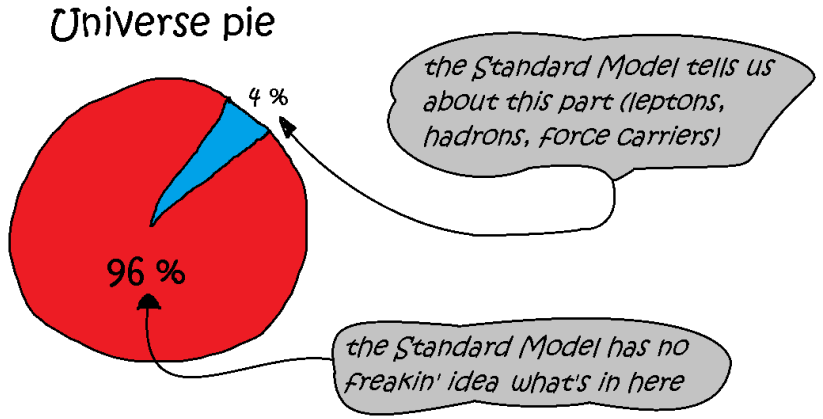
"I've either discovered dark matter,
or I've left the lens cap on."

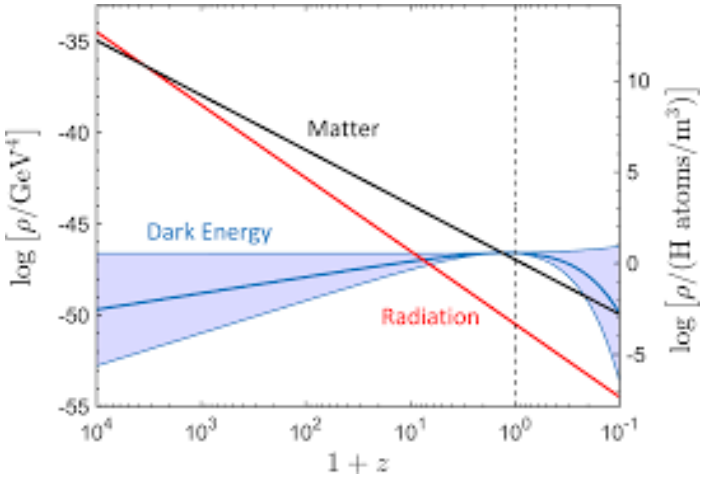


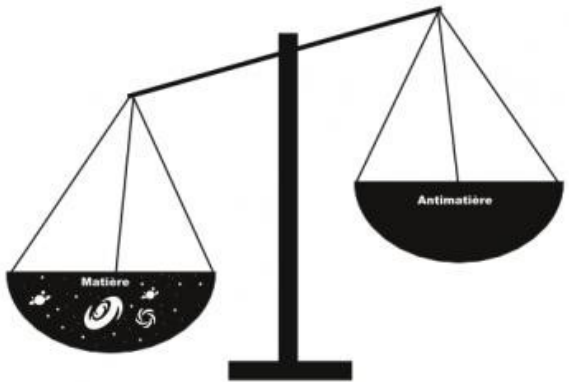


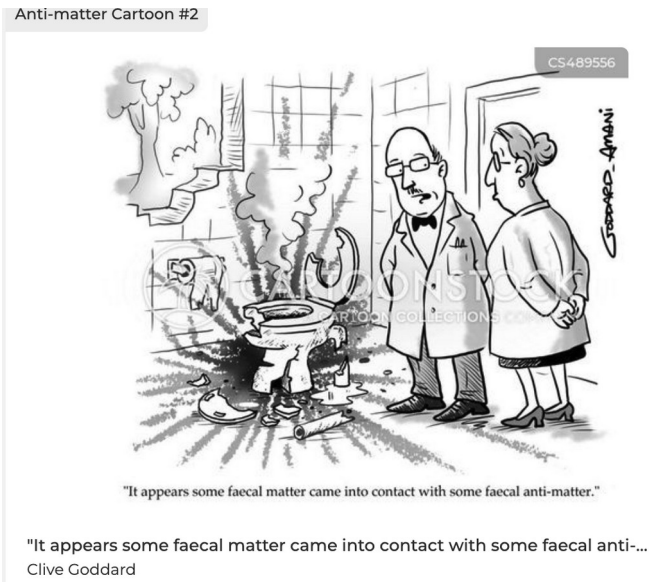


WHY NEW PHYSICS (ALL ABOUT STATISTICS DOES UNIVERSE PLAY WITH DICE)?



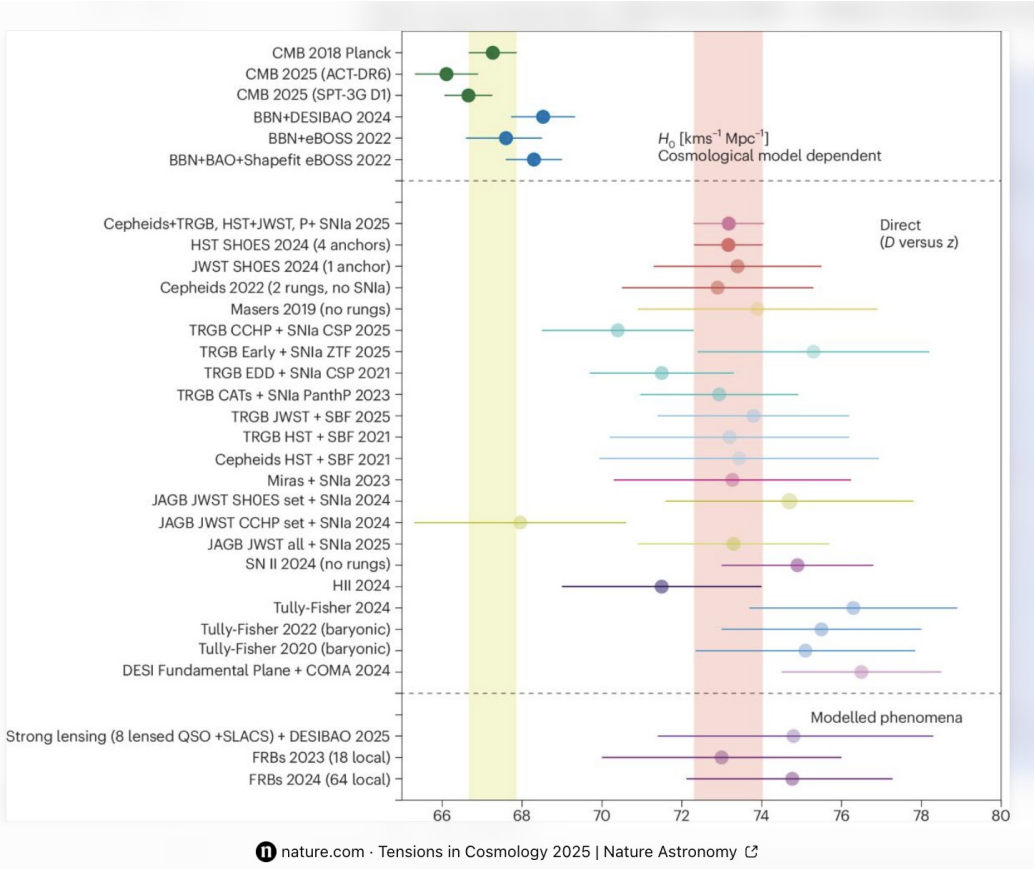


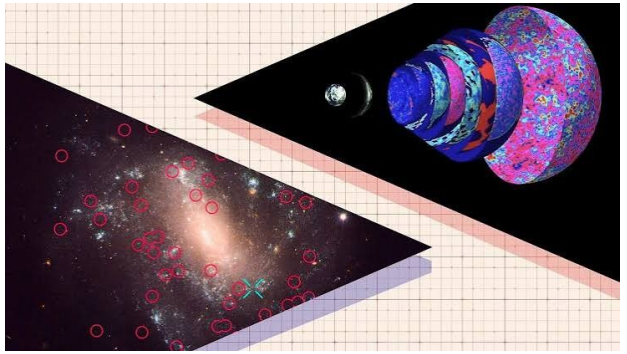


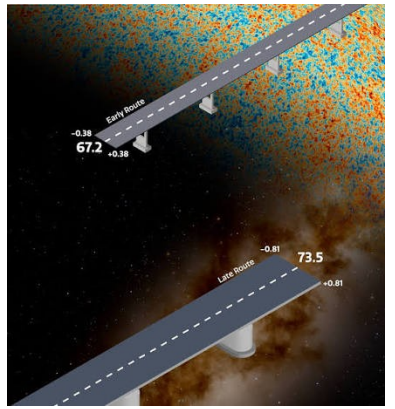


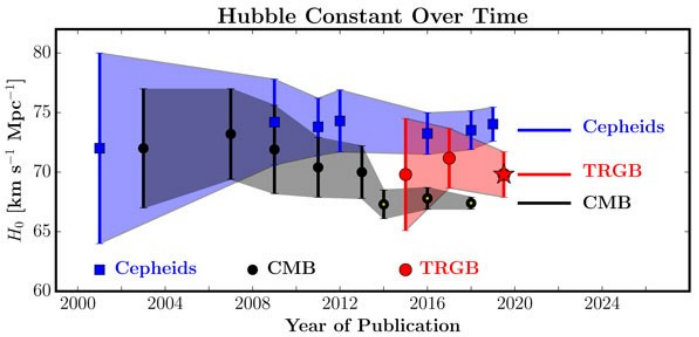
NEW PHYSICS: (WHEN THE PRESENT MEETS THE PAST)

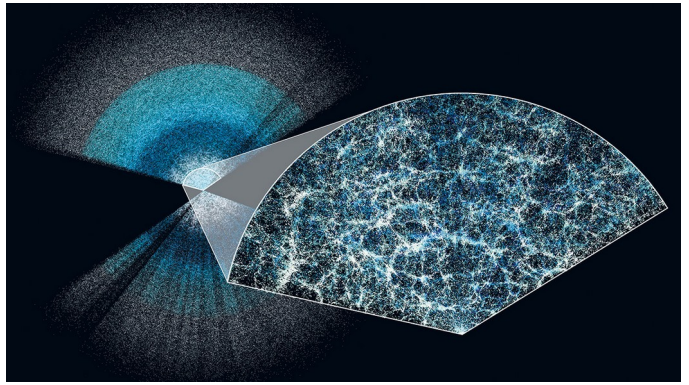
CHILDREN CONFRONT THEIR PARENTS….





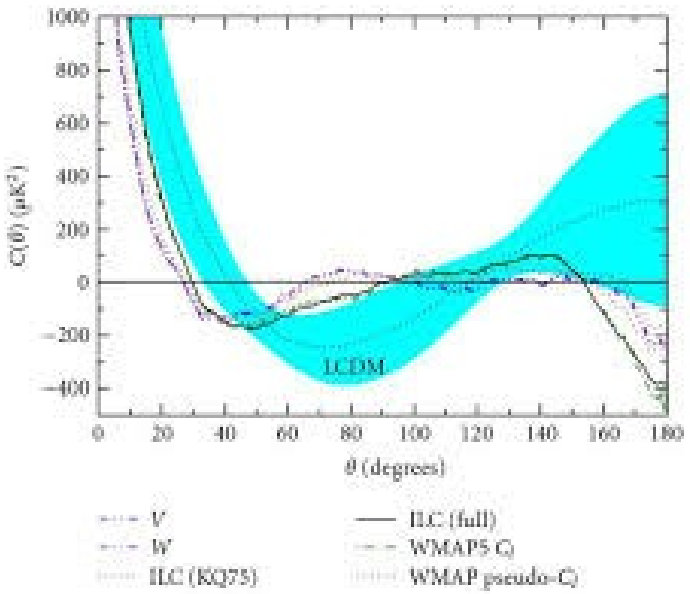


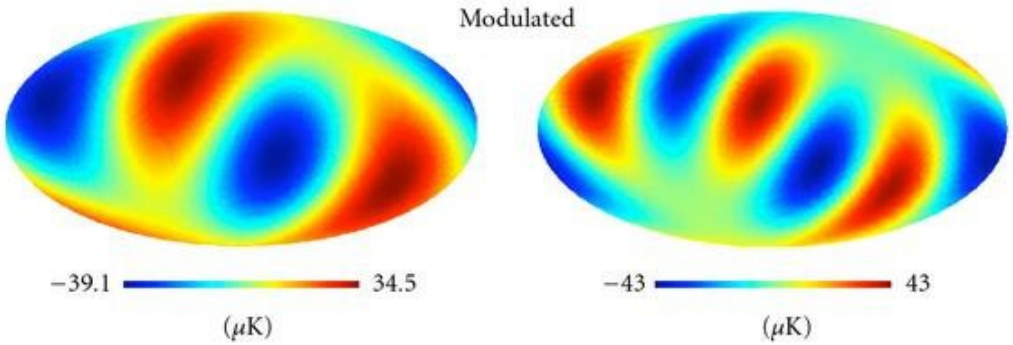


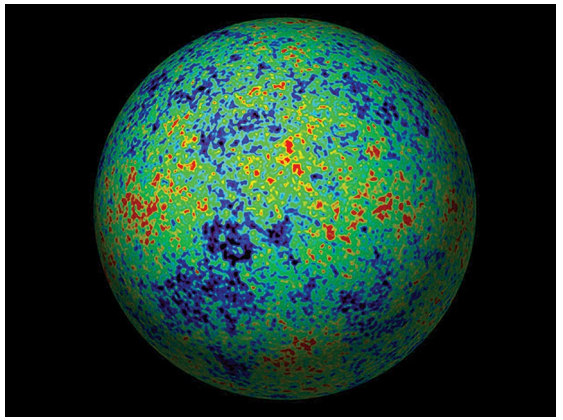


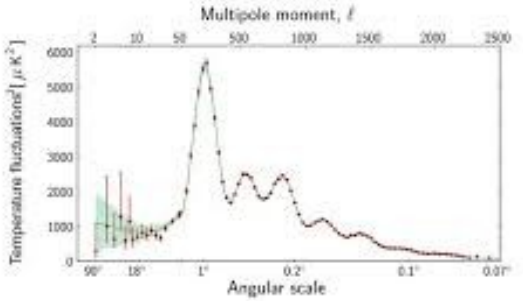
DESI's map of the universe is the largest to date, showing delicate bubble-like structures in the distribution of galaxies (inset) that contain clues to the expansion history of the universe. Credit: C Lamman/DESI/cmastro

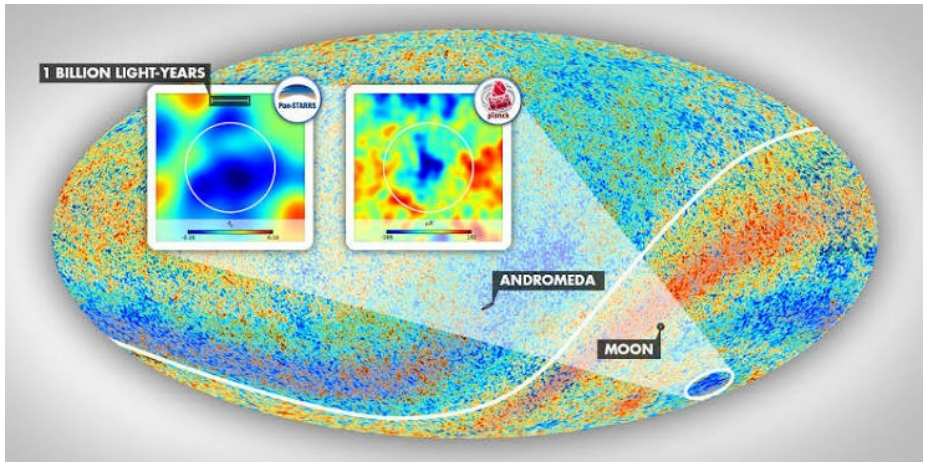
NEW PHYSICS (CMB ANOMALIES) FANTOMS OR REALITY





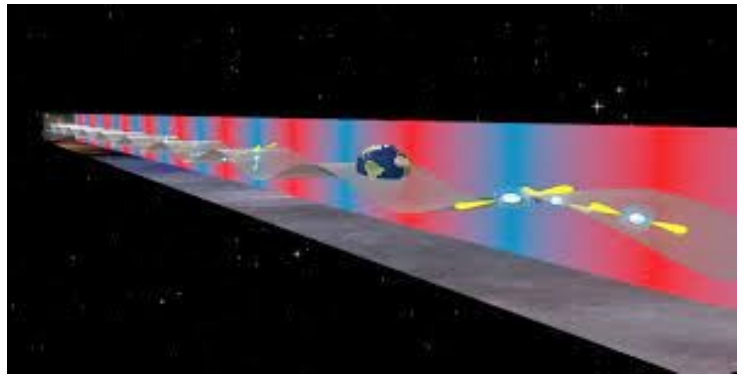




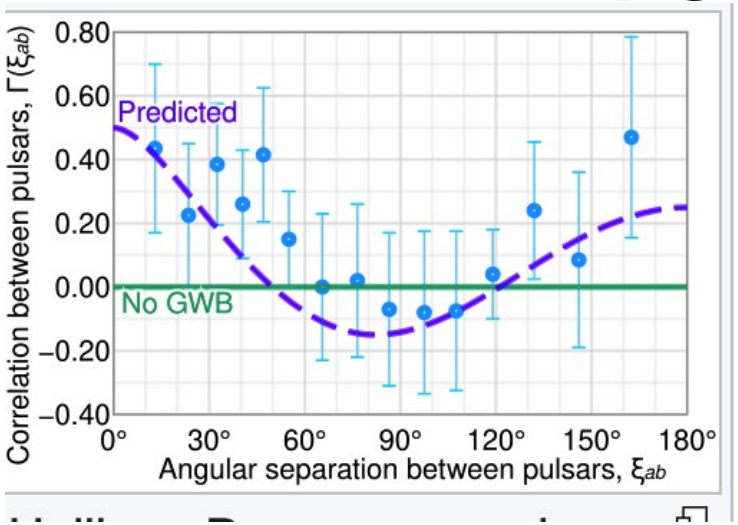


Cold cosmic mystery solved: Largest known structure in the universe leaves its imprint on CMB radiation
by University of Hawaii at Manoa

PULSAR TIMING ARRAYS



The proposal to use pulsars as GW detectors was originally made by **Sazhin (1978)** and **Detweiler (1979)** iThe idea is to treat the solar system barycenter and a galactic pulsar as opposite ends of an imaginary arm in space.



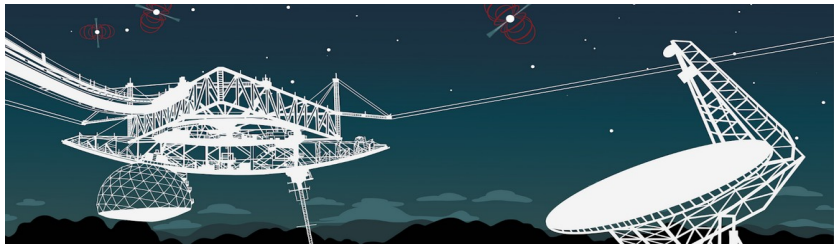
$$\Gamma_{ab} = \frac{1}{2}\delta_{ab} + \frac{1}{2} - \frac{x_{ab}}{4} + \frac{3}{2}x_{ab}\ln x_{ab}$$

$$x_{ab} = (1 - \cos\zeta_{ab})/2$$

overlap reduction function

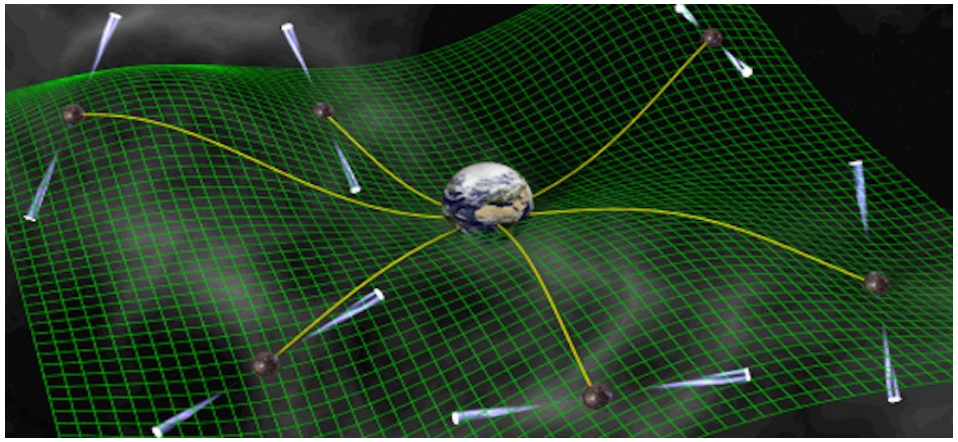
Credit: NANOGrav

In 1983, **Hellings** and **Downs** extended this idea to an array of pulsars and found that a stochastic background of GWs would produce a distinctive GW signature: a quadrupolar spatial correlation between arrival times of pulses emitted by different millisecond pulsar pairings that depends only on the pairing's angular separation in the sky as viewed from Earth (actually the solar system barycenter). The key property of a pulsar timing array is that the signal from a stochastic GW background will be correlated across the sightlines of pulsar pairs, while that from the other noise processes will not.



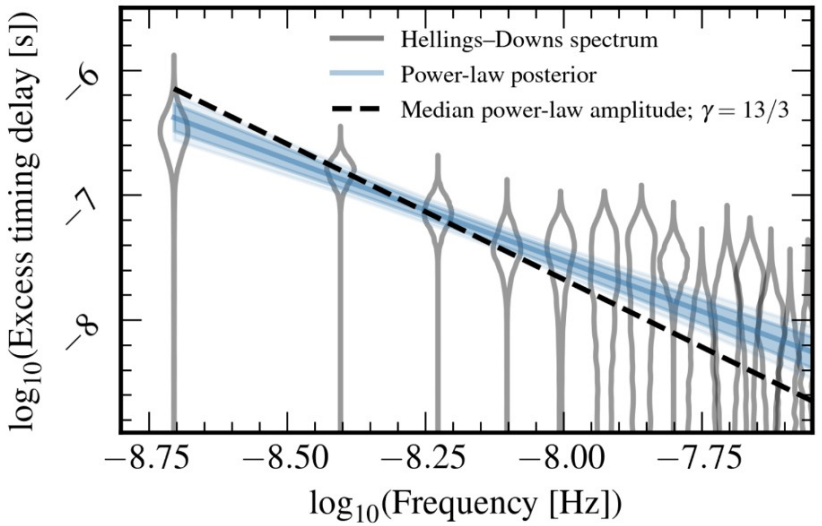
Assumptions: Isotropic GW that obeys Einstein's general relativity. It is valid for "long-arm" detectors like pulsar timing arrays, where the wavelengths of typical GWs are much shorter than the "long-arm" distance between Earth and typical pulsars

STOCHASTIC GRAVITATIONAL WAVES DETECTION



The pulsar acts as the reference clock at one end of the arm sending out regular signals which are monitored by an observer on Earth. The effect of a passing long-wavelength GW would be to perturb the galactic spacetime and cause a small change in the observed time of arrival of the pulses.

Credit: NANOGrav

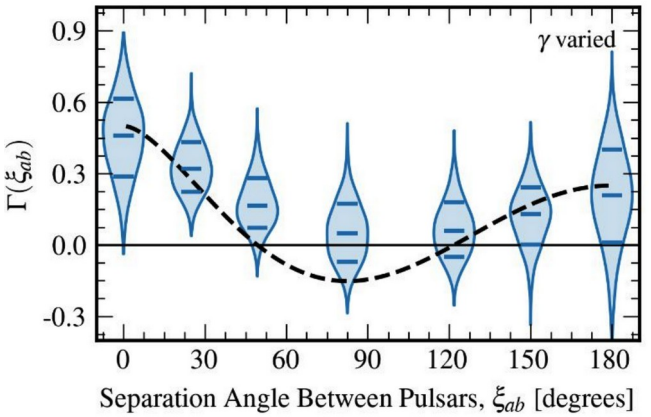


Detection residuals in the time delay in the pulsar signals

$$R(t) \equiv \int_0^t dt' \left(\frac{\nu_0 - \nu(t')}{\nu_0} \right)$$

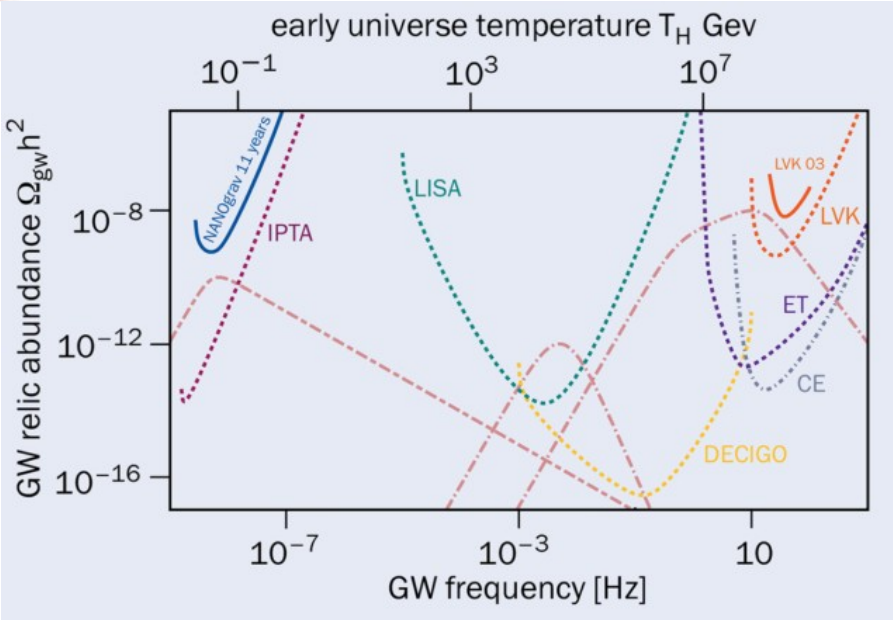
$$\frac{1}{T_{\rm obs}} < f < \frac{1}{\delta t}$$

$$h_c(f) = A_{\rm CP} \left(\frac{f}{f_{\rm yr}} \right)^{\alpha_{\rm CP}},$$

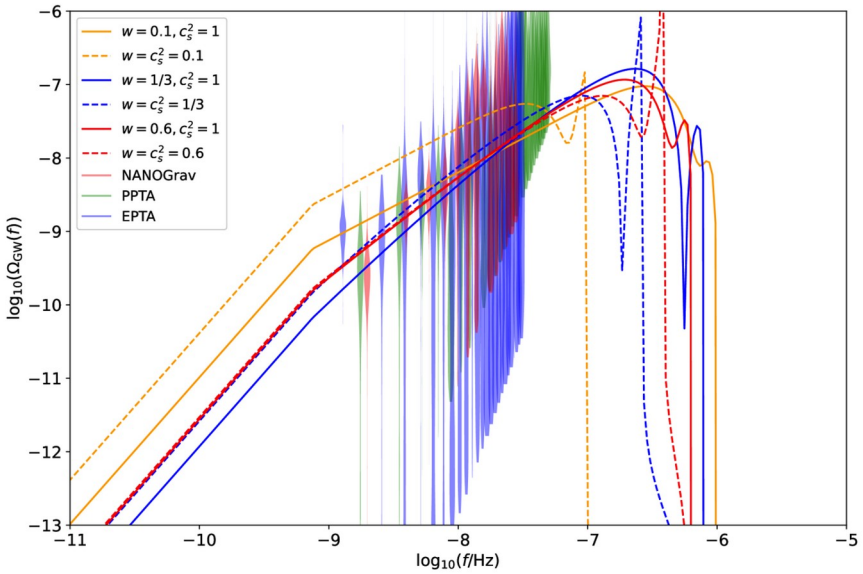


$$\Omega_{\rm GW}(f) = \frac{2\pi^2}{3H_0^2} f^2 h_c^2(f) = \Omega_{\rm GW}^{\rm yr} \left(\frac{f}{f_{\rm yr}} \right)^{5-\gamma_{\rm CP}}$$

SIGNAL FROM EARLY UNIVERSE?



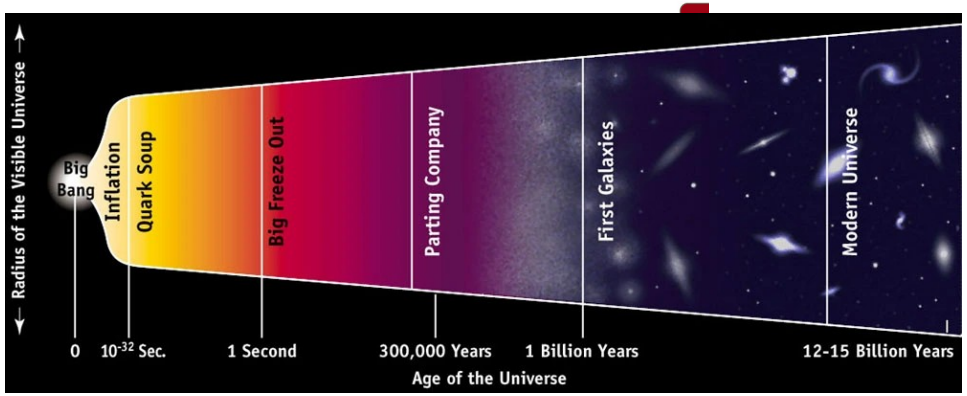
Credit: CERN Courier, 2021



Credit: Liu, Wu, Chen

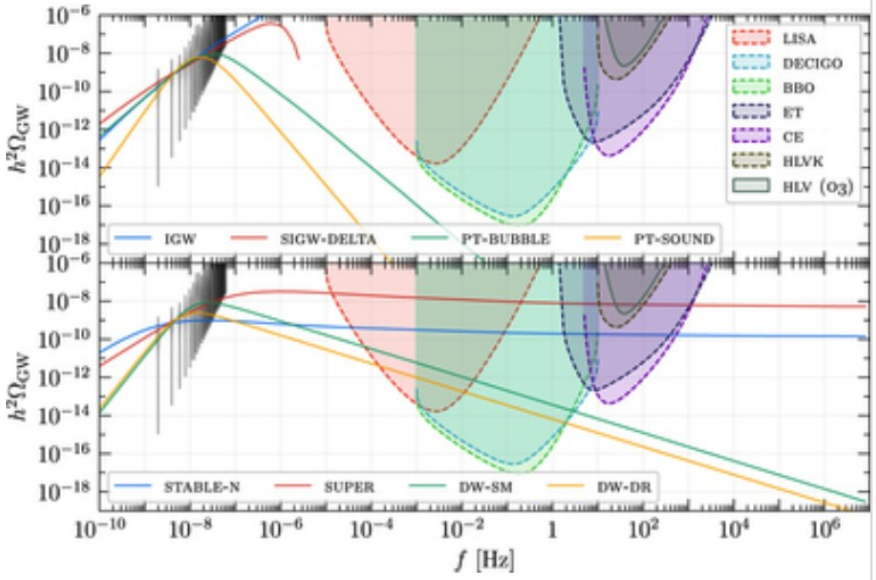
Figure 1: The energy density spectrum of SIGW with different EoS parameter w and sound speed c_s as a function of GW frequency.

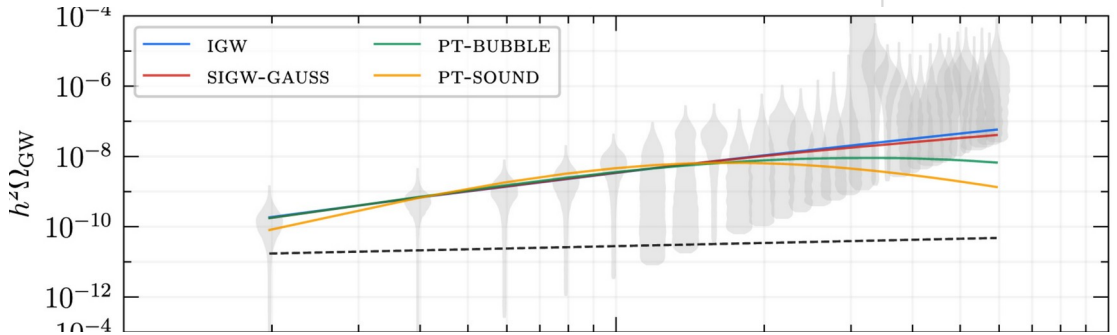
Sensitivity of current (solid) and future (dashed) gravitational-wave (GW) observatories to stochastic GW backgrounds (expressed in terms of the energy density fraction in the universe today). On the upper x-axis, the temperature in the early universe is given, which is obtained when the peak frequency of a GW signal is equal to the inverse of the expansion rate when GWs are emitted. Some example possible GW spectra from the early universe are also shown (pink, dashed). F. Rompineve/ arXiv:2101.12130/arXiv:2002.0461



PULSAR TIMING ARRAYS AND NEW PHYSICS

NANOGrav 2023





Figueroa et al. 2024

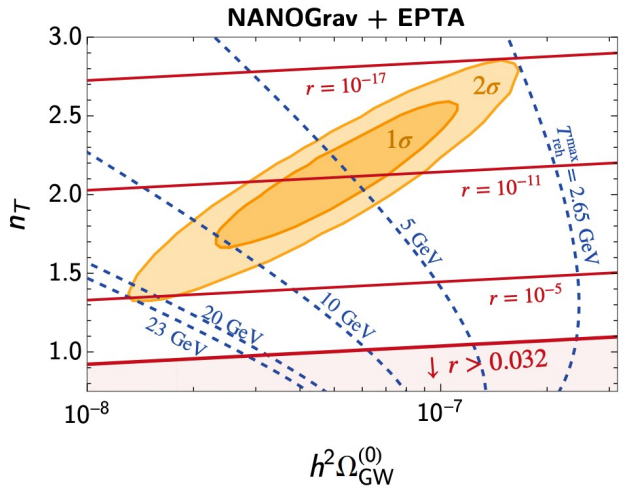


FIG. S.1. The CMB observations tightly constrain the PTA-signal interpretation as the inflationary GWB (blue contours). The bound on the tensor-to-scalar ratio (in red) excludes the region with $r > 0.032$. The BBN-$N_{\rm eff}$ bound requires a low-scale reheating: $T_{\rm reh} < 10.6$ GeV, at 68% CL.

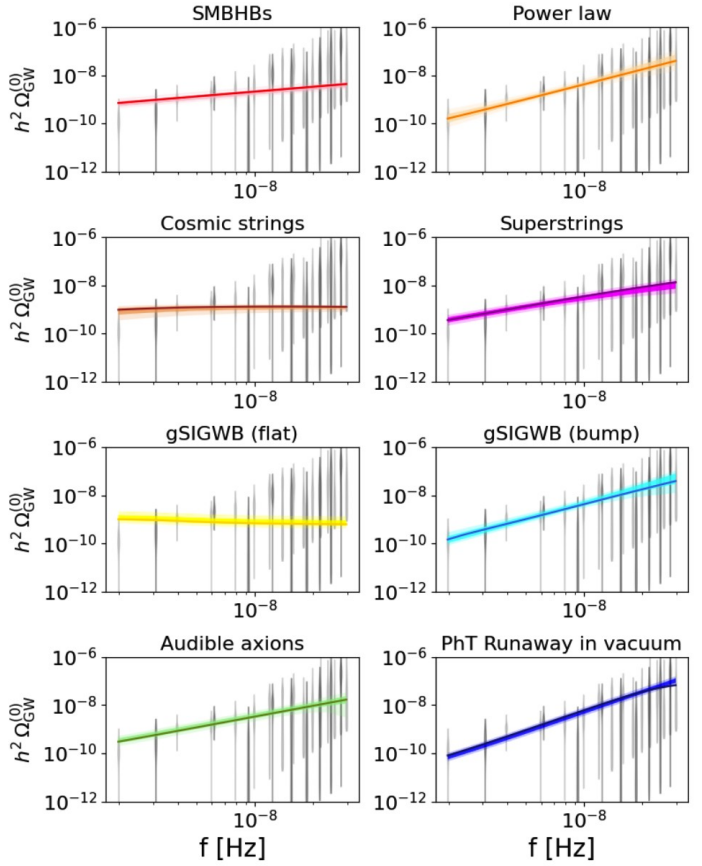


FIG. 2. $h^2\Omega_{\rm GW}^{(0)}(f)$ for various cases. Solid-colored lines represent central values, while lighter (lightest) regions the 68% (95%) CL intervals. Dark and light gray violins are the EPTA and NANOGrav free spectrum data, respectively.

15-YR DATA NANOGRAV

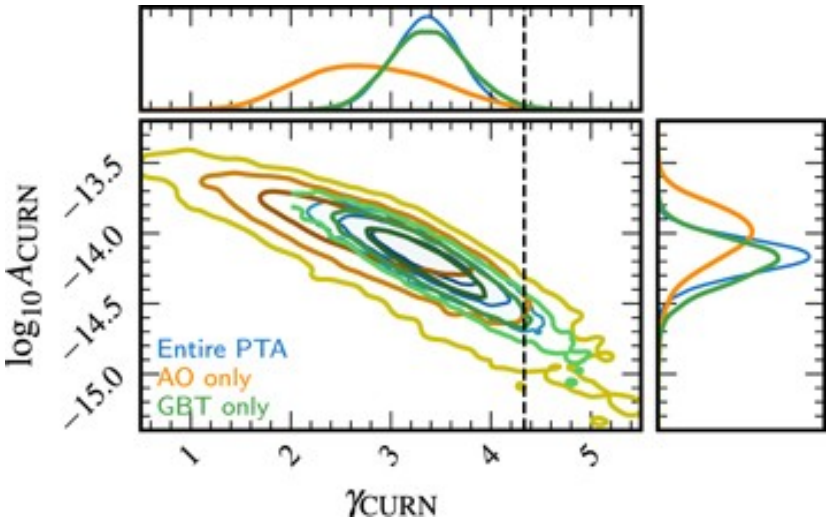
Astrophysical:

- ✓ Super massive black hole binary (SMBHB)

(Phinney 2001): γ=13/3

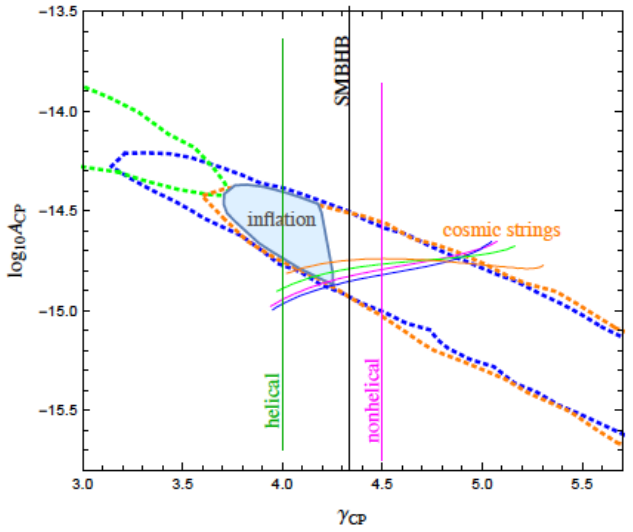
Cosmological:

- ✓ Bubbles collisions (Kosowsky et. al. 1993)
- ✓ Inflation (Vagnozzi 2020)
- ✓ Cosmic strings (Blanco-Pillado et al. 2020)
- ✓ Sound Waves (Hindmarsh et al. 2014)
- ✓ Seed magnetic fields (Neronov et. al. 2020)
- ✓ Hydrodynamic and MHD Turbulence (Brandenburg et al. 2021)



The NANOGrav 15 yr Data Set: Detection of a Gravitational-wave Background

$$h_c(f) = A_{\rm CP}\left(\frac{f}{f_{\rm yr}}\right)^{\alpha_{\rm CP}},$$

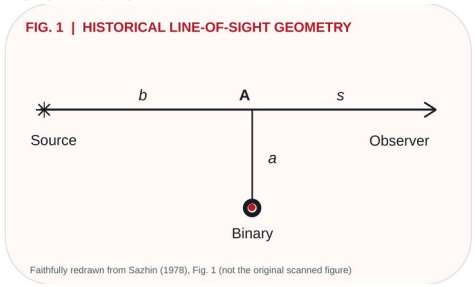


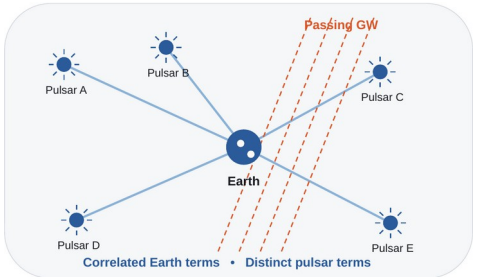
Credit: Emma Clarke

QCD PHASE TRANSITIONS GRAVITATIONAL WAVES

Sazhin, 1978, Detweiler 1979

Pulsar Timing Arrays (PTAs) are sensible to gravitational waves generated or present at QCD energy scales





- A gravitational wave changes the propagation time of electromagnetic pulses.
- Stable pulsars can serve as clocks for testing ultralong-period gravitational waves.
- The original figure depicts a binary system near the radio-signal trajectory.

CONCEPTUAL PROGRESSION

1978 — **Sazhin**
Pulse propagation time

1979 — **Detweiler**
Timing residuals and GW background

1983 — **Hellings & Downs**
Angular cross-correlations

TODAY — **PTAs**
Nanohertz GW constraints

HISTORICAL ESTIMATE

Temperature
T ≈ 300 MeV

Present-day frequency
$\nu_0 \approx 40$ nHz

Present-day strain estimate
$h_0 \approx 3 \times 10^{-15}$

Within the PTA frequency band

From one cosmic clock to a GW observatory

Deryagin, Grigoriev, Rubakov & Sazhin, Mon. Not. R. Astron. Soc. 229, 357–370 (1987),

ORIGINAL PAPER EXCERPT

Smaller anisotropic domains may appear during the QCD phase at temperatures above $T_g = 300\,\mathrm{MeV}$. The amplitude of the anisotropic pressure is hard to calculate although dimensional estimates (Shaposhnikov 1987) give $\Delta \sim 10^{-2}-10^{-3}$. In this case the gravitational wave background is also generated, and the present amplitude and the frequency is of order of

$$h_0 = 3\times10^{-15}, \qquad \nu_0 = 4\times10^{-8}\,\mathrm{Hz}.$$

To conclude this section we mention that the electroweak and QCD phase transitions could also have produced the gravitational waves (Witten 1984; Hogan 1986). The full sketch of the generation mechanisms and the possibilities of the detection can be found in the review by Thorne (1987).

COSMOLOGICAL PHASE TRANSITIONS: BUBBLES

Kosowsky, et al, 1992a,b & 1993, Kamionkowsky et al. 1994

VOLUME 69, NUMBER 14 PHYSICAL REVIEW LETTERS 5 OCTOBER 1992

Gravitational Waves from First-Order Cosmological Phase Transitions

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(Received 6 December 1991; revised manuscript received 26 May 1992)

A first-order cosmological phase transition that proceeds through the nucleation and collision of true-vacuum bubbles is a potent source of gravitational radiation. Possibilities for such include first-order inflation, grand-unified-theory-symmetry breaking, and electroweak-symmetry breaking. We have calculated gravity-wave production from the collision of two scalar-field vacuum bubbles, and, using an approximation based upon these results, from the collision of 20 to 30 vacuum bubbles. We present estimates of the relic background of gravitational waves produced by a first-order phase transition; in general, $\Omega_{GW} \sim 10^{-9}$ and $f \sim (10^{-6}\ \mathrm{Hz})(T/1\ \mathrm{GeV})$.

PHYSICAL REVIEW D VOLUME 47, NUMBER 10 15 MAY 1993

Gravitational radiation from colliding vacuum bubbles: Envelope approximation to many-bubble collisions

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(Received 9 November 1992)

We introduce an approximation to calculate the gravitational radiation produced by the collision of true-vacuum bubbles that is simple enough to allow the simulation of a phase transition by the collision of hundreds of bubbles. This "envelope approximation" neglects the complicated "overlap" regions of colliding bubbles and follows only the evolution of the bubble walls. The approximation accurately reproduces previous results for the gravitational radiation from the collision of two scalar-field vacuum bubbles. Using a bubble nucleation rate given by $\Gamma = \Gamma_0 e^{\beta t}$, we simulate a phase transition by colliding 20 to 200 bubbles; the fraction of vacuum energy released into gravity waves is $E_{GW}/E_{vac} = 0.06(H/\beta)^2$ and the peak of the spectrum occurs at $\omega_{max} = 1.6\beta$ ($H^2 = 8\pi G\rho/3$ is the Hubble constant associated with the false-vacuum phase). The spectrum is very similar to that in the two-bubble case, except that the efficiency of gravity-wave generation is about five times higher, presumably due to the fact that a given bubble collides with many others. Finally, we consider two further "statistical" approximations, where the gravitational radiation is computed as an incoherent sum over individual bubbles weighted by the distribution of bubble sizes. These approximations provide reasonable estimates of the gravitational-wave spectrum with far less computation.

- T∗ – the Universe temperature at which the PT takes place.
- α∗ – the strength of the phase transition, defined as the ratio of the vacuum and relativistic energy density at PT
- β/H∗ – the bubble nucleation rate in units of the Hubble rate at the time of the PT, H∗.
- v – the velocity of the bubble walls.

PHYSICAL REVIEW D VOLUME 45, NUMBER 12 15 JUNE 1992

Gravitational radiation from colliding vacuum bubbles

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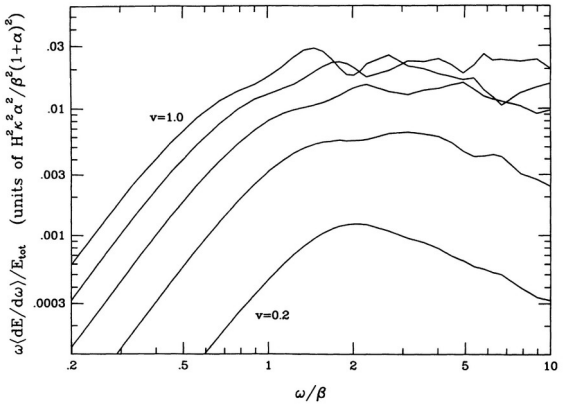
In the linearized-gravity approximation we numerically compute the amount of gravitational radiation produced by the collision of two true-vacuum bubbles in Minkowski space. The bubbles are separated by distance d and we calculate the amount of gravitational radiation that is produced in a time $\tau \sim d$ (in a cosmological phase transition τ corresponds to the duration of the transition, which is expected to be of the order of the mean bubble separation d). Our approximations are generally valid for $\tau \lesssim H^{-1}$. We find that the amount of gravitational radiation produced depends only upon the grossest features of the collision: the time τ and the energy density associated with the false-vacuum state, ρ_{vac}. In particular, the spectrum $dE_{GW}/d\omega \propto \rho_{vac}^2 \tau^8$ and peaks at a characteristic frequency $\omega_{max} \simeq 3.8/\tau$, and the fraction of the vacuum energy released into gravitational waves is about $1.3 \times 10^{-3} (\tau/H^{-1})^2$, where $H^2 = 8\pi G\rho_{vac}/3$ (τ/H^{-1} is expected to be of the order of a few percent). We address in some detail the important symmetry issues in the problem, and how the familiar "quadrupole approximation" breaks down in a most unusual way: it *overestimates* the amount of gravitational radiation produced in this highly relativistic situation by more than a factor of 50. Most of our results are for collisions of bubbles of equal size, though we briefly consider the collision of vacuum bubbles of unequal size. Our work implies that the vacuum-bubble collisions associated with a strongly first-order phase transition are a very potent cosmological source of gravitational radiation.



The energy per octave radiated in gravity waves for a phase transition with spherical bubbles expanding at velocity v, for v = 0.2, v = 0.4, v = 0.6, v = 0.8, and v = 1.0.

Pioneering works:

- *Winicour 1973*
- *Hogan 1982, 1986*
- *Turner & Wilczek 1990*
- *Kosowsky, Turner, Watkins. 1992*

$$\Omega_{\mathrm{GW}} h^2 \approx 1.1 \times 10^{-6} \kappa^2 \left(\frac{H_*}{\beta}\right)^2 \left(\frac{\alpha}{1+\alpha}\right)^2 \left(\frac{v^3}{0.24+v^3}\right) \left(\frac{100}{g_*}\right)^{1/3},$$

$$f_{\mathrm{max}} \approx 5.2 \times 10^{-8}\,\mathrm{Hz} \left(\frac{\beta}{H_*}\right) \left(\frac{T_*}{1\,\mathrm{GeV}}\right) \left(\frac{g_*}{100}\right)^{1/6},$$

$$h_c(f_{\mathrm{max}}) \approx 1.8 \times 10^{-14} \kappa \left(\frac{\alpha}{1+\alpha}\right) \left(\frac{H_*}{\beta}\right)^2 \left(\frac{1\,\mathrm{GeV}}{T_*}\right) \left(\frac{v^3}{0.24+v^3}\right)^{1/2} \left(\frac{100}{g_*}\right)^{1/3}.$$

For follow papers see: Caprini and Figueroa 2018

COSMOLOGICAL PHASE TRANSITIONS: HYDRO- AND MHD TURBULENCE

Kosowsky, et al, 2002

PHYSICAL REVIEW D 66, 024030 (2002)

Gravitational radiation from cosmological turbulence

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(Received 6 April 2002; published 24 July 2002)

An injection of energy into the early Universe on a given characteristic length scale will result in turbulent motions of the primordial plasma. We calculate the stochastic background of gravitational radiation arising from a period of cosmological turbulence, using a simple model of isotropic Kolmogoroff turbulence produced in a cosmological phase transition. We also derive the gravitational radiation generated by magnetic fields arising from a dynamo operating during the period of turbulence. The resulting gravitational radiation background has a maximum amplitude comparable to the radiation background from the collision of bubbles in a first-order phase transition, but at a lower frequency, while the radiation from the induced magnetic fields is always subdominant to that from the turbulence itself. We briefly discuss the detectability of such a signal.

$$\nabla^2 h_{ij}(\boldsymbol{x},t) - \frac{\partial^2}{\partial t^2} h_{ij}(\boldsymbol{x},t) = -16\pi G\, S_{ij}(\boldsymbol{x},t)$$

$$\gamma H_*^{-1} = 2\pi/k_0, \qquad \zeta H_*^{-1} = \tau_T; \qquad \mathrm{w} = \frac{4\rho_*}{3} = \frac{2\pi^2}{45} g_* T_*^4, \qquad H_* = 1.66 g_*^{1/2} \frac{T_*^2}{m_{\mathrm{Pl}}},$$

$$= 1.55 \times 10^{-3}\ \mathrm{Hz} \left(\frac{\omega_*}{k_0}\right)\left(\frac{g_*}{100}\right)^{1/6}\left(\frac{\gamma}{0.01}\right)^{-1}\left(\frac{T_*}{100\ \mathrm{GeV}}\right),$$

PHYSICAL REVIEW D VOLUME 49, NUMBER 6 15 MARCH 1994

Gravitational radiation from first-order phase transitions

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(Received 26 October 1993)

We consider the stochastic background of gravity waves produced by first-order cosmological phase transitions from two types of sources: colliding bubbles and hydrodynamic turbulence. First we discuss the fluid mechanics of relativistic spherical combustion. We then numerically collide many bubbles expanding at a velocity v and calculate the resulting spectrum of gravitational radiation in the linearized gravity approximation. Our results are expressed as simple functions of the mean bubble separation, the bubble expansion velocity, the latent heat, and the efficiency of converting latent heat to kinetic energy of the bubble walls. A first-order phase transition is also likely to excite a Kolmogoroff spectrum of turbulence. We estimate the gravity waves produced by such a spectrum of turbulence and find that the characteristic amplitude of the gravity waves produced is comparable to that from bubble collisions. Finally, we apply these results to the electroweak transition. Using the one-loop effective potential for the minimal electroweak model, the characteristic amplitude of the gravity waves produced is $h \simeq 1.5 \times 10^{-27}$ at a characteristic frequency of 4.1×10^{-3} Hz corresponding to $\Omega \sim 10^{-22}$ in gravity waves, far too small for detection. Gravity waves from more strongly first-order phase transitions, including the electroweak transition in nonminimal models, have better prospects for detection, though probably not by LIGO.

PACS number(s): 04.30.Db, 98.70.Vc, 98.80.Cq

COSMOLOGICAL PHASE TRANSITIONS: SOUND WAVES

Hindmarsh et al. 2014, 2015

PRL 112, 041301 (2014) PHYSICAL REVIEW LETTERS week ending 31 JANUARY 2014

Gravitational Waves from the Sound of a First Order Phase Transition

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We report on the first three-dimensional numerical simulations of first-order phase transitions in the early Universe to include the cosmic fluid as well as the scalar field order parameter. We calculate the gravitational wave (GW) spectrum resulting from the nucleation, expansion, and collision of bubbles of the low-temperature phase, for phase transition strengths and bubble wall velocities covering many cases of interest. We find that the compression waves in the fluid continue to be a source of GWs long after the bubbles have merged, a new effect not taken properly into account in previous modeling of the GW source. For a wide range of models, the main source of the GWs produced by a phase transition is, therefore, the sound the bubbles make.

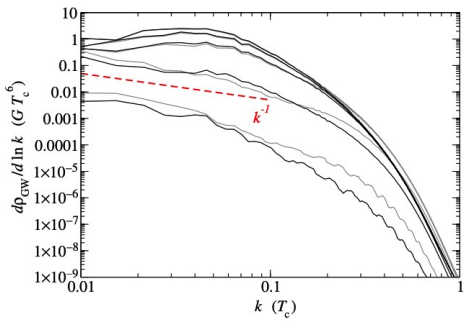

FIG. 3 (color online). Gravitational wave power spectra during the phase transition, for the intermediate strength transition, from fluid only (black lines) and both fluid and field (gray lines). From bottom to top, the times are $t = 600$, 800, 1000, 1200, and $1400T_c^{-1}$. The red dashed line indicates the expected k^{-1} behavior.

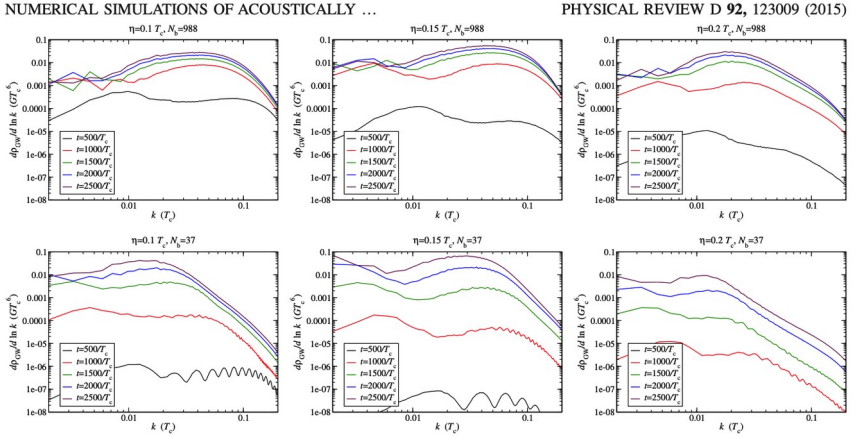

FIG. 8 (color online). Gravitational wave power spectra, for weak transitions, at $\eta/T_c = 0.1$, 0.15 and 0.2 ($v_w = 0.83$, 0.54 and 0.44) for $N_b = 988$ (top row) and $N_b = 37$ (bottom row). Note that the axes and time intervals are the same for all plots, which means that in some cases the latest $(2500/T_c)$ curve is from before the completion of the phase transition.

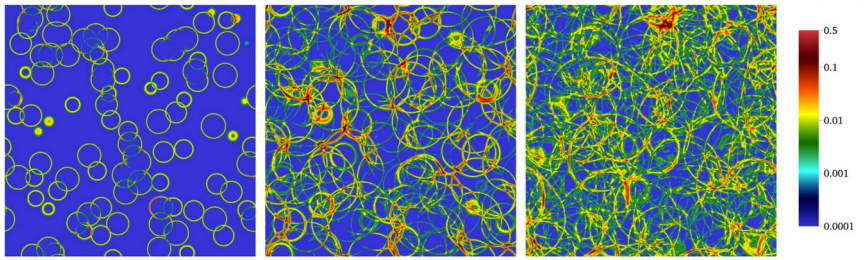

FIG. 4 (color online). Slices of fluid kinetic energy density E/T_c^4 at $t = 500T_c^{-1}$, $t = 1000T_c^{-1}$ and $t = 1500T_c^{-1}$ respectively, for the $\eta/T_c = 0.15$, $N_b = 988$ simulation.

PHYSICAL REVIEW D 92, 123009 (2015)

Numerical simulations of acoustically generated gravitational waves at a first order phase transition

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(Received 5 May 2015; published 22 December 2015)

We present details of numerical simulations of the gravitational radiation produced by a first order thermal phase transition in the early Universe. We confirm that the dominant source of gravitational waves is sound waves generated by the expanding bubbles of the low-temperature phase. We demonstrate that the sound waves have a power spectrum with a power-law form between the scales set by the average bubble separation (which sets the length scale of the fluid flow L_f) and the bubble wall width. The sound waves generate gravitational waves whose power spectrum also has a power-law form, at a rate proportional to L_f and the square of the fluid kinetic energy density. We identify a dimensionless parameter $\tilde{\Omega}_{GW}$ characterizing the efficiency of this "acoustic" gravitational wave production whose value is $8\pi\tilde{\Omega}_{GW} \simeq 0.8 \pm 0.1$ across all our simulations. We compare the acoustic gravitational waves with the standard prediction from the envelope approximation. Not only is the power spectrum steeper (apart from an initial transient) but the gravitational wave energy density is generically larger by the ratio of the Hubble time to the phase transition duration, which can be 2 orders of magnitude or more in a typical first order electroweak phase transition.

For follow papers see: Caprini and Figueroa 2018

PULSAR TIMING ARRAYS: COSMOLOGICAL PHASE TRANSITIONS

PHYSICAL REVIEW LETTERS **127,** 251302 (2021)

Editors' Suggestion | Featured in Physics

Searching for Gravitational Waves from Cosmological Phase Transitions with the NANOGrav 12.5-Year Dataset

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(NANOGrav Collaboration)

We search for a first-order phase transition gravitational wave signal in 45 pulsars from the NANOGrav 12.5-year dataset. We find that the data can be modeled in terms of a strong first order phase transition taking place at temperatures below the electroweak scale. However, we do not observe any strong preference for a phase-transition interpretation of the signal over the standard astrophysical interpretation in terms of supermassive black hole mergers; but we expect to gain additional discriminating power with future datasets, improving the signal to noise ratio and extending the sensitivity window to lower frequencies. An interesting open question is how well gravitational wave observatories could separate such signals.

TABLE III. Prior distributions for the parameters used in all the analyses in this work. The prior for the bubble wall velocity reported in this table is the one used for the SWO analysis, for the BO analyses we use $v_w = 1$ as explained in the text.

Parameter	Description	Prior	Comments
	White noise		
E_k	EFAC per backend-receiver system	Uniform [0, 10]	Single-pulsar analysis only
Q_k [s]	EQUAD per backend-receiver system	Log-uniform $[-8.5, -5]$	Single-pulsar analysis only
J_k [s]	ECORR per backend-receiver system	Log-uniform $[-8.5, -5]$	Single-pulsar analysis only
	Red noise		
$A_{\rm red}$	Red-noise power-law amplitude	Log-uniform $[-20, -11]$	One parameter per pulsar
$\gamma_{\rm red}$	Red-noise power-law spectral index	Uniform [0, 7]	One parameter per pulsar
	Phase transition		
T_* [GeV]	Phase transition temperature	Log-uniform $[-4, 3]$	One parameter for PTA
α_*	Phase transition strength	Log-uniform $[-1.3, 1]$	One parameter for PTA
H_*/β	Bubble nucleation rate	Log-uniform $[-2, 0]$	One parameter for PTA
v_w	Bubble wall velocity	Log-uniform $[-2, 1]$	One parameter for PTA
	Supermassive black bole binaries (SMBHBs)		
$A_{\rm GWB}$	Common process strain amplitude	Log-uniform $[-18, -14]$	One parameter for PTA
$\gamma_{\rm GWB}$	Common process power-law spectral index	Delta function ($\gamma_{\rm GWB} = 13/3$)	Fixed

TABLE I. Parameters for the gravitational wave spectrum of Eq. (4). The values of the parameters (a, b, c) in the spectral shape of the bubble contribution are reported in Table II.

	Bubbles [58]	Sound waves [59]
$\Delta(v_w)$	$[0.48 v_w^3/(1 + 5.3 v_w^2 + 5 v_w^4)]$	$0.513 v_w$
κ	κ_ϕ	$\kappa_{\rm SW}$
p	2	2
q	2	1
$\mathcal{S}(x)$	$\{(a+b)^c/[b x^{-a/c} + a x^{b/c}]^c\}$	$x^3[7/(4+3x^2)]^{7/2}$
f_*/β	$[0.35/(1 + 0.07 v_w + 0.69 v_w^4)]$	$(0.536/v_w)$

$$h^2\Omega(f) = \mathcal{R}\Delta(v_w)\left(\frac{\kappa\alpha_*}{1+\alpha_*}\right)^p\left(\frac{H_*}{\beta}\right)^q\mathcal{S}(f/f_*^0),$$

$$f_*^0 \simeq 1.13\times10^{-10}\ \text{Hz}\left(\frac{f_*}{\beta}\right)\left(\frac{\beta}{H_*}\right)\left(\frac{T_*}{\text{MeV}}\right)\left(\frac{g_*}{10}\right)^{1/6},$$

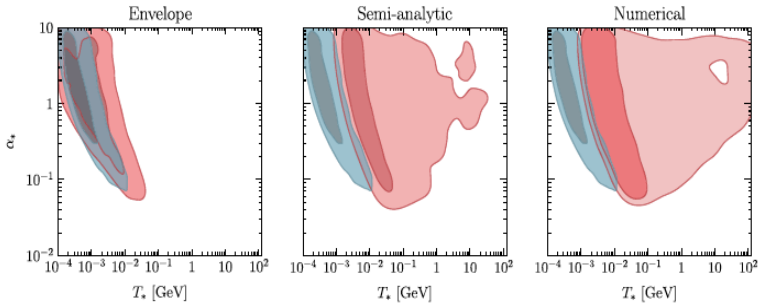


FIG. 1. In red (blue) the 1-σ (68% posterior credible level), and 2-σ (95% posterior credible level) contours for the two-dimensional posterior distributions in the (T_*, α_*) plane obtained in the BO (SWO). The BO analysis has been performed with the spectral shape computed by using the envelope approximation (left panel), semianalytic results (central panel), and numerical results (right panel). Specifically, we use $(a, b, c) = (1, 2.61, 1.5)$ for the semianalytic results, and $(a, b, c) = (0.7, 2.3, 1)$ for the numerical results.

TABLE II. Comparison of the bubble spectral shape parameters derived using the envelope and thin wall approximation [58] (left column), the semianalytic approach of reference [64] (middle column), and lattice simulations [62] (right column). For numerical and semianalytic results the values of the parameters depend on the choice of the scalar field potential; we report the range of values obtained for the different scalar field potentials considered in the above mentioned works.

	Envelope	Semianalytic	Numerical
a	3	1–2.2	1.6–0.7
b	1	2.6–2.9	1.4–2.3
c	1.5	1.5–3.5	1
f_*/β	$[0.35/(1 + 0.07 v_w + 0.69 v_w^2)]$	0.1	0.2

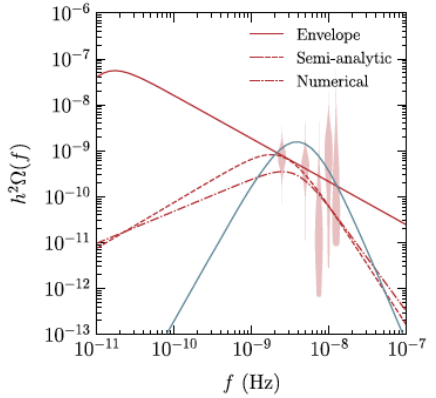
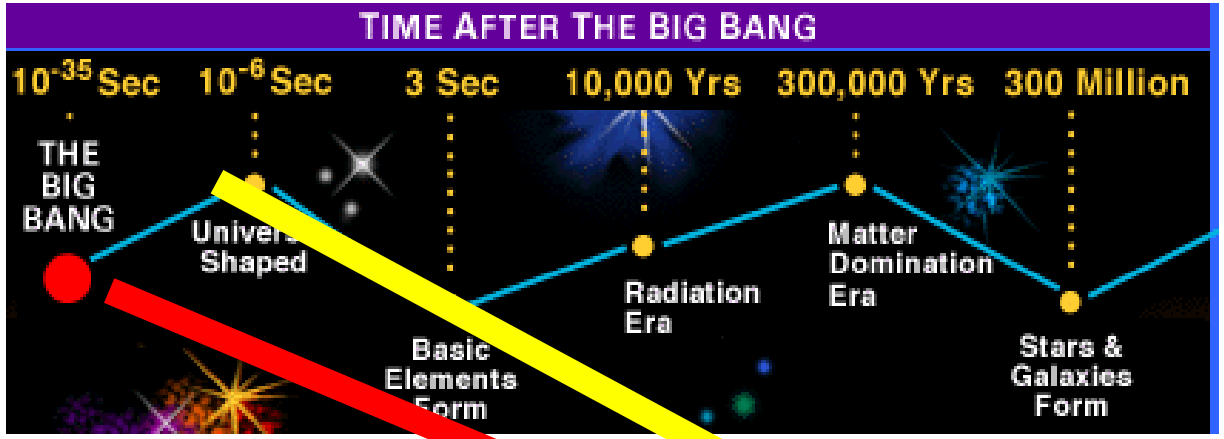
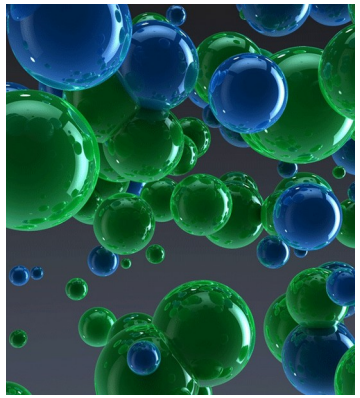


FIG. 2. Maximum likelihood GWB fractional energy-density spectrum for the BO (red) and SWO (blue) analyses compared with the marginalized posterior for the free power spectrum (independent per-frequency characterization; red violin plot) derived in NG12gwb For the BO analysis we show the results derived by using the envelope (solid line), semianalytic (dashed), and numerical (dot-dashed) spectral shapes. For the BO analyses the values of (α_*, T_*) for these maximum likelihood spectra are (0.28,0.7 MeV) for the envelope results, (1.2,3.4 MeV) for the semianalytic results, and (0.13,14.1 MeV) for the numerical results. While for the SO analysis we get (6.0,0.32 MeV).

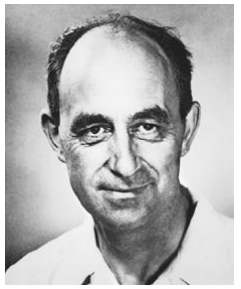
WHY PRIMORDIAL TURBULENCE?



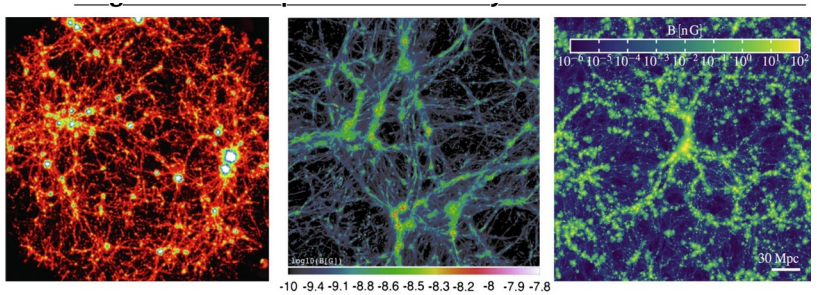


magnetic field origin

- red-inflation
- yellow- phase transitions

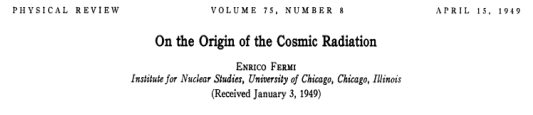


F. Hoyle
in Proc. "*La structure et l'evolution de l'Universe*" (1958)



Projection of magnetic field strength in three cosmological simulations using different MHD approaches and solvers. Left: non-radiative GADGET SPH simulations with galactic seeding by Donnert et al. (2009), based on the MHD method by Dolag and Stasyszyn (2009). Middle: non-radiative ENZO MHD simulation on a fixed grid by Vazza et al. (2014) using the Dender cleaning (Dedner et al. 2002); Right: Simulation with full "Illustris TNG" galaxy formation model using a Lagrangian finite volume method (Marinacci et al. 2018b)

Donnert et al. 2018

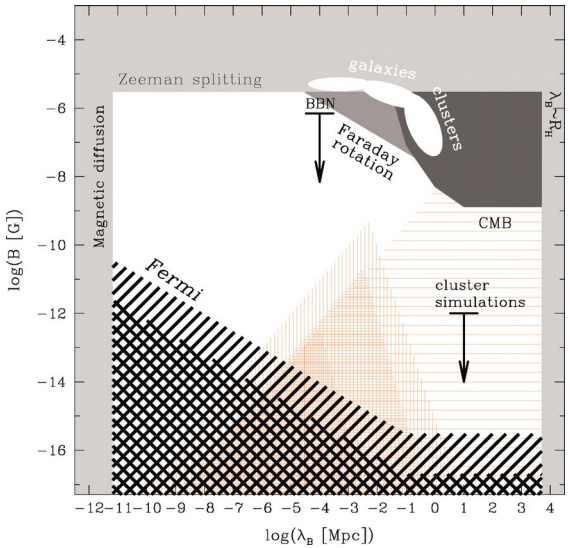


PHYSICAL REVIEW VOLUME 75, NUMBER 8 APRIL 15, 1949

On the Origin of the Cosmic Radiation

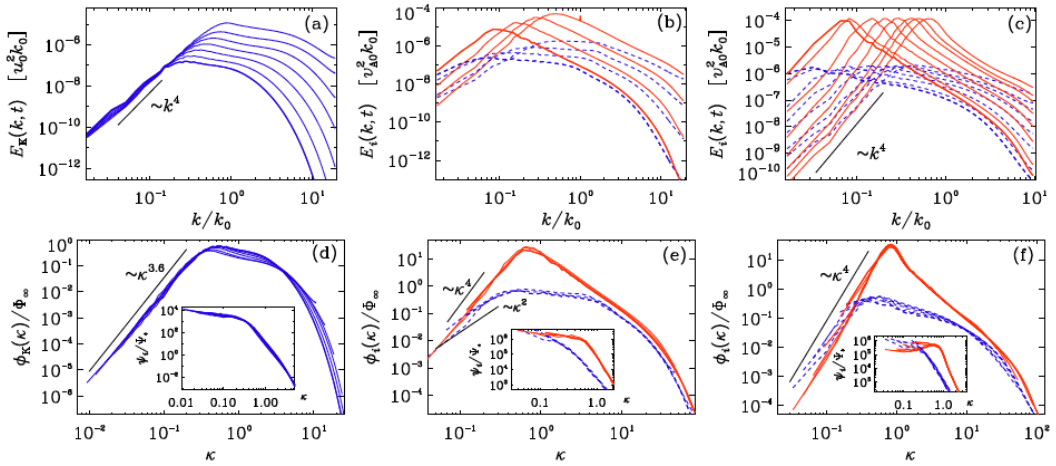
ENRICO FERMI
Institute for Nuclear Studies, University of Chicago, Chicago, Illinois
(Received January 3, 1949)

A theory of the origin of cosmic radiation is proposed according to which cosmic rays are originated and accelerated primarily in the interstellar space of the galaxy by collisions against moving magnetic fields. One of the features of the theory is that it yields naturally an inverse power law for the spectral distribution of the cosmic rays. The chief difficulty is that it fails to explain in a straightforward way the heavy nuclei observed in the primary radiation.

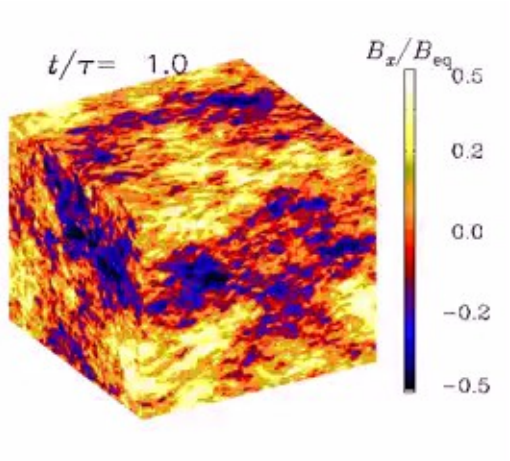


A. Neronov & E. Vovk," *Evidence for Strong Extra-galactic Magnetic Fields from Fermi Observations of TeV Blazars*", Science 328, 5974 (2010)

HYDRO – AND MHD TURBULENCE GRAVITATIONAL WAVES



Brandenburg and Kahniashvili 2017



$$\frac{\partial \ln\rho}{\partial t} = -\frac{4}{3}(\boldsymbol{\nabla}\cdot\boldsymbol{u} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\ln\rho) + \frac{1}{\rho}[\boldsymbol{u}\cdot(\boldsymbol{J}\times\boldsymbol{B}) + \eta\boldsymbol{J}^2],$$

$$\frac{\partial \boldsymbol{u}}{\partial t} = -\boldsymbol{u}\cdot\boldsymbol{\nabla}\boldsymbol{u} + \frac{\boldsymbol{u}}{3}(\boldsymbol{\nabla}\cdot\boldsymbol{u} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\ln\rho) - \frac{\boldsymbol{u}}{\rho}[\boldsymbol{u}\cdot(\boldsymbol{J}\times\boldsymbol{B}) + \eta\boldsymbol{J}^2] - \frac{1}{4}\boldsymbol{\nabla}\ln\rho + \frac{3}{4\rho}\boldsymbol{J}\times\boldsymbol{B} + \frac{2}{\rho}\boldsymbol{\nabla}\cdot(\rho\nu\boldsymbol{\mathsf{S}}) + \boldsymbol{\mathcal{F}},$$

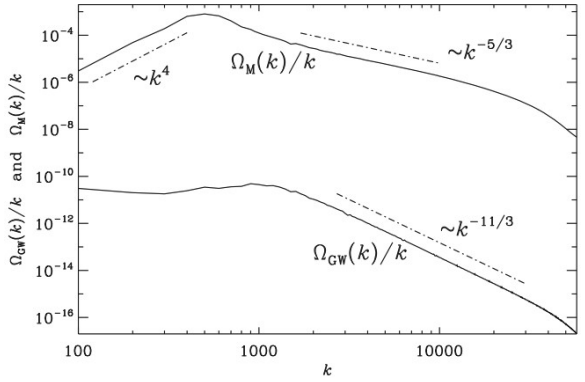


FIG. 1. Magnetic and GW energy spectra for run ini2 averaged over late times ($t > 1.1$), after the GW spectrum has started to fluctuate around a steady state.

$$(\partial_t^2 - c^2\boldsymbol{\nabla}^2)h_{ij}(\boldsymbol{x},t) = \frac{16\pi G}{ac^2}T_{ij}^{\rm TT}(\boldsymbol{x},t)$$

$$T_{ij} = \frac{4}{3}\frac{\rho u_i u_j}{1-\boldsymbol{u}^2} - B_i B_j + (\rho/3 + \boldsymbol{B}^2/2)\delta_{ij},$$

TABLE II. Correspondence between the slopes expected from Ref. [12] for the subinertial range ("ana") and what is obtained in our run ini2 ("sim"), and the results for spectra with the Kolmogorov slope ("Kol") and the Golitsyn slope ("Gol"), which agrees with Ref. [12].

Slope of	ana	sim	Kol	Gol
$\Omega_{\rm M}$	5	5	−2/3	−8/3
$\Omega_{\rm GW}$	3	1	−8/3	−14/3
h_c	1/2	−1/2	−7/3	−10/3

Roper Pol et al.. 2019

PHYSICAL REVIEW D **102,** 083512 (2020)

Numerical simulations of gravitational waves from early-universe turbulence

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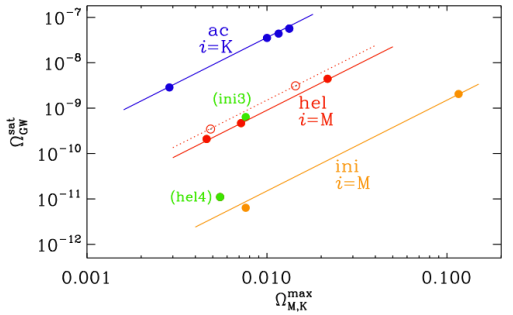
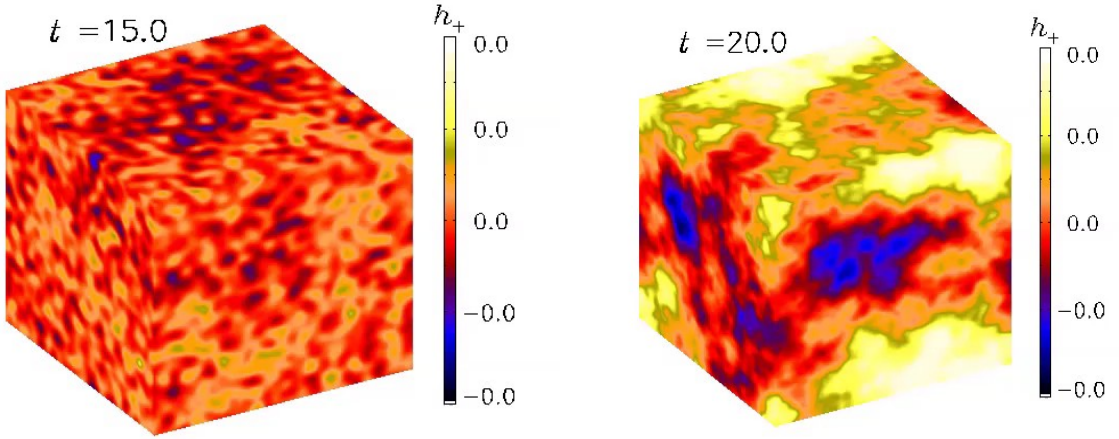


FIG. 7. $\Omega_{\rm GW}^{\rm sat}$ versus $\Omega_{\rm M,K}^{\rm max}$. The quadratic dependence inferred from the +2 slope of the lines holds within runs of the same type. Note that runs ini3 ($N = 10$) and hel4 ($N = 1000$) in green have different stirring scales than the rest of the runs ($N = 100$).

HYDRO – AND MHD TURBULENCE GRAVITATIONAL WAVES

Credit: Axel Brandenburg



$$\mathcal{E}_M(t) \simeq \mathrm{w} b_1^2 \left(1+\frac{t}{\tau_1}\right)^{-2/3},$$

$$\mathcal{E}_v(t) \simeq \mathrm{w} v_1^2 \left(1+\frac{t}{\tau_1}\right)^{-2/3},$$

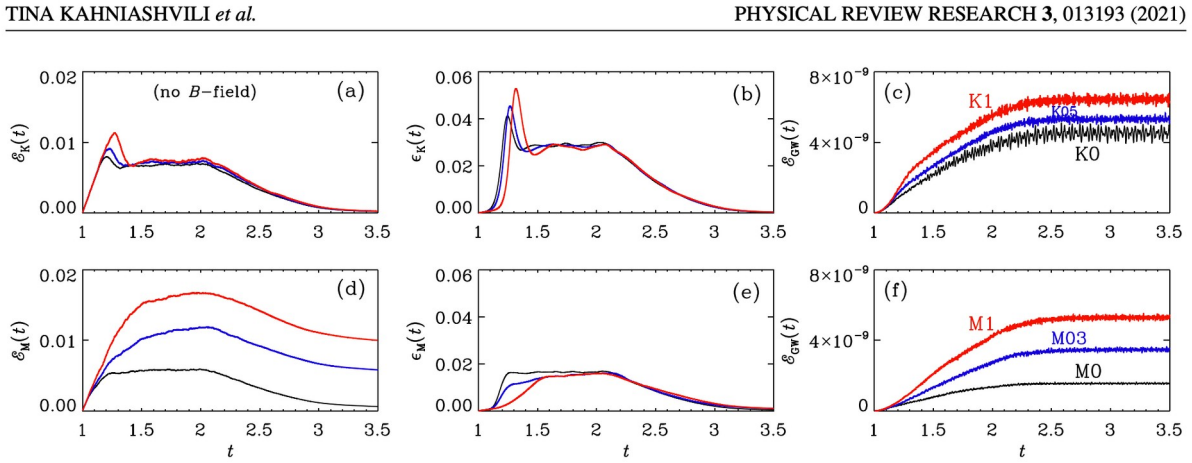


FIG. 2. Evolution of (a) $\mathcal{E}_K$, (b) ϵ_K, and (c) $\mathcal{E}_{GW}$ for kinetically driven cases with $\sigma = 0$ (run K0, black), 0.5 (run K05, blue), and 1 (run K1, red), and of (d) $\mathcal{E}_M$, (e) ϵ_M, and (f) $\mathcal{E}_{GW}$ for magnetically driven cases with $\sigma = 0$ (run M0, black), 0.3 (run M03, blue), and 1 (run M1, red).

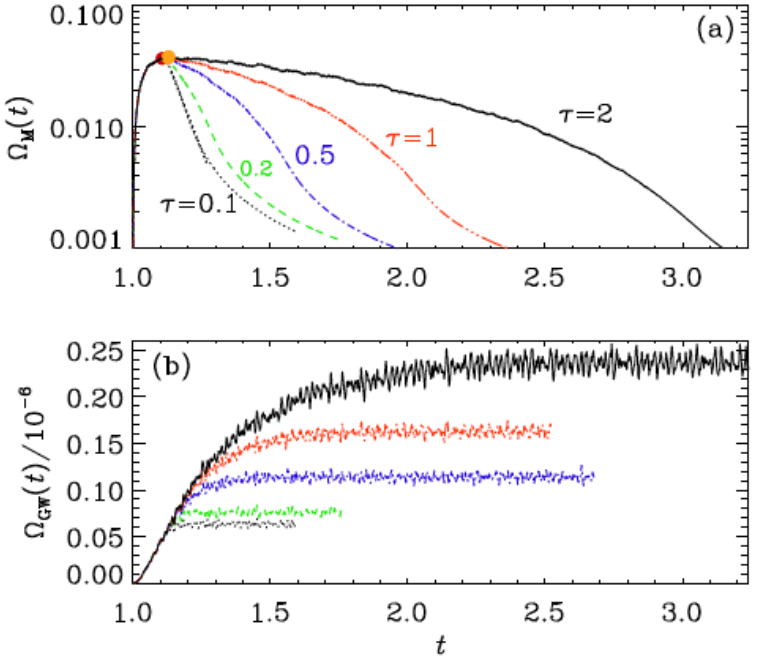


FIG. 1: Evolution of magnetic energy (top) and growth of GW energy density (bottom) for simulations where the driving is turned off at $t = 1.1$ (black dotted line), or the strength of the driving is reduced linearly in time over the duration $\tau = 0.2$ (green), 0.5 (blue), 1 (red), or 2 (black). Time is in units of the Hubble time at the moment of source activation.

Kahniashvili et al. 2022

PRIMORDIAL MAGNETIC FIELDS LIMITS FROM BBN

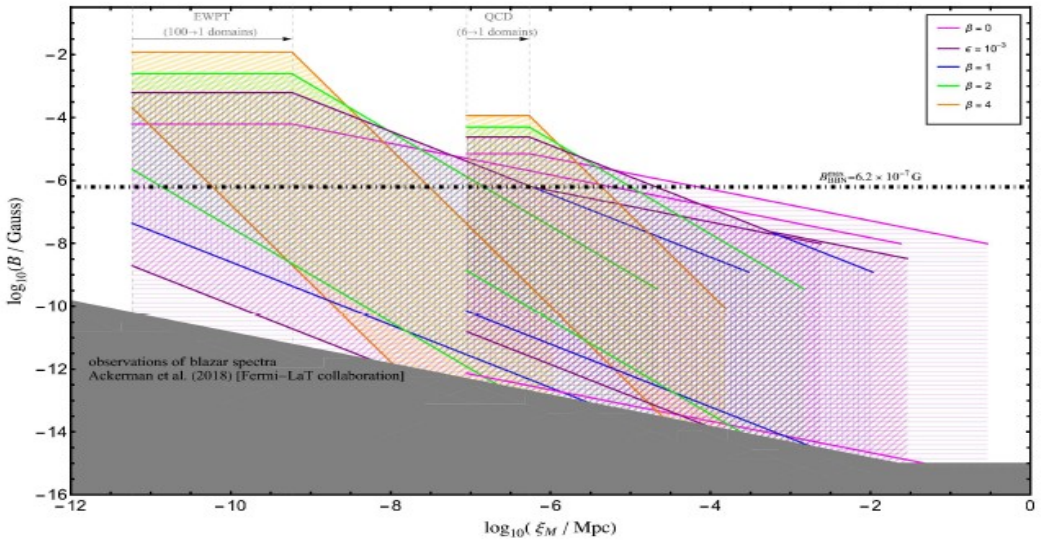
Kaahniashvili, et al. 2022

- EXTRA RADIATION LIKE ENERGY DENSITY LESS THAN ~3% OF THE RADIATION ENERGY DENSITY AT BBN

$$\frac{\rho_{\rm add}}{\rho_{\rm rad}} = 0.277\left(\frac{\Delta N_{\rm eff}}{0.122}\right); \qquad \Delta N_{eff} = N_{\rm eff} - N^{\nu}_{\rm eff}$$

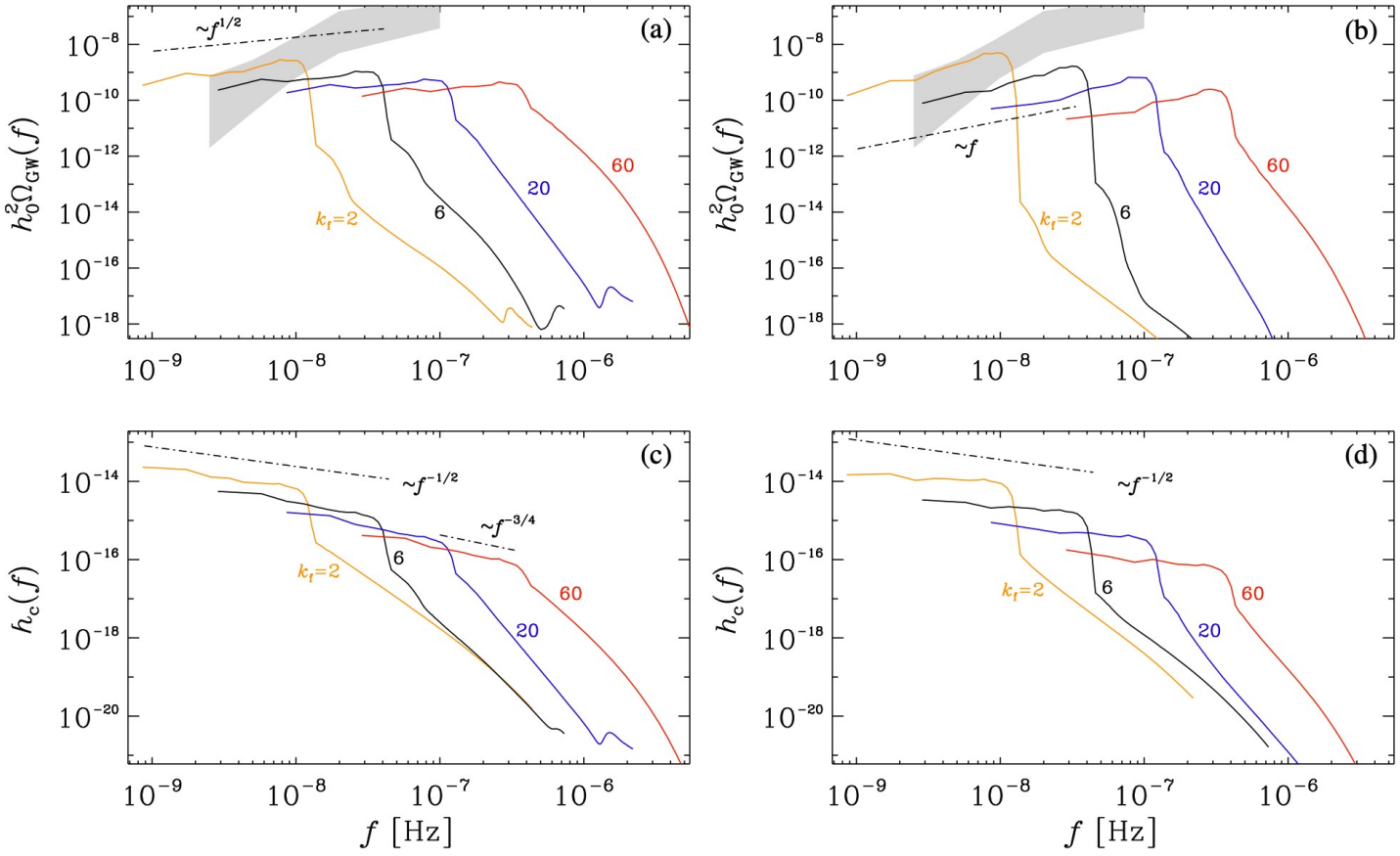
- THE UPPER BOUND ON THE MAGNETIC (EFFECTIVE) AMPLITUDE ORDER OF MICROGAUSS AT BBN
- ACCOUNTING FOR THE MAGNETIC FIELD DECAY:
  - THE MAGNETIC ENERGY DENSITY DOES NOT EXCEED THE RADIATION ENERGY DENSITY AT THE MOMENT OF GENERATION
  - BBN BOUNDS ARE SATISFIED

Possible turbulent evolution of the comoving MF strength B (and correlation length ξ_M from generation at the EW and QCD scales in the cases of fully helical (β = 0), nonhelical (β = 1, 2, 4), and partially helical MHD turbulence. Upper limits on ξ_M are determined by the size of the horizon and number of domains (bubbles) at generation, ranging from 1 to 6 (at QCD) or 100 (at EW), depending on the PT modeling. Lines terminate (on the right) at recombination (T = 0.25 eV). The upper limit of the comoving MF strength at BBN (T = 0.1 MeV) is indicated by the black dot-dashed line. Regimes excluded by observations of blazar spectra are marked in gray. The hatched regions are bounded by an (upper) limit from BBN and a (lower) limit from the blazar spectra.,



GRAVITATIONAL WAVES FROM PRIMORDIAL HD & MHD TURBULENCE

Kaahniashvili, et al. 2022



$$\mathcal{E}_{GW} = \left(q \frac{\mathcal{E}_M}{k_f} \right)^n$$

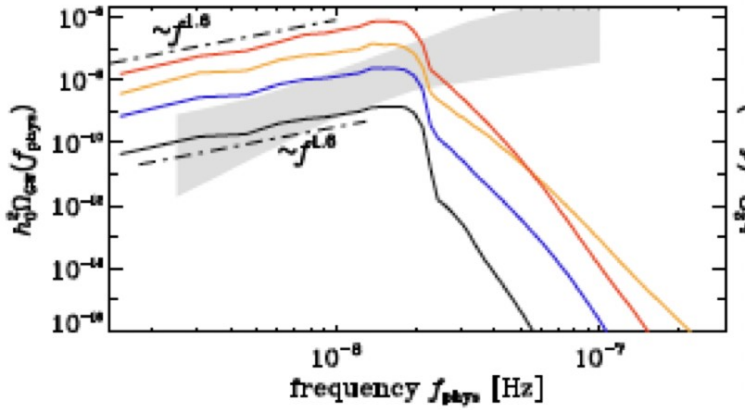


FIG. 2: Frequency spectra, $h_0^2\Omega_{GW}(f)$, for both the QCDPT R
orange, blue, and black, respectively.

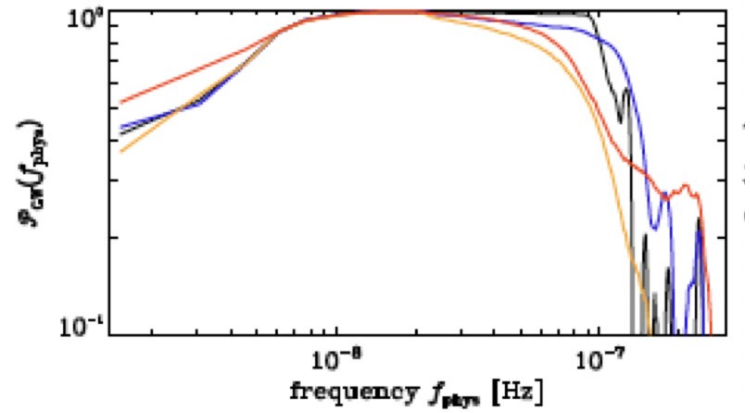


FIG. 3: Polarization spectra, $\mathcal{P}_{GW}(f)$, for the QCDPT Runs a
orange, blue, and black, respectively.

PULSAR TIMING ARRAYS AND EARLY UNIVERSE TURBULENCE

A&A, 685, A94 (2024)
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Astronomy & Astrophysics

The second data release from the European Pulsar Timing Array

IV. Implications for massive black holes, dark matter, and the early Universe★

EPTA Collaboration and InPTA Collaboration: J. Antoniadis (Ιωάννης Αντωνιάδης)[1,2], P. Arumugam[3], S. Arumugam[4], S. Babak[6], M. Bagchi[7,8], A.-S. Bak Nielsen[2,9], C. G. Bassa[11], A. Bathula[12], A. Berthereau[13,14], M. Bonetti[15,16,17], E. Bortolas[15,16,17], P. R. Brook[18], M. Burgay[19], R. N. Caballero[20], A. Chalumeau[15], D. J. Champion[2], S. Chanlaridis[1], S. Chen[23], I. Cognard[13,14], S. Dandapat[24], D. Deb[7], S. Desai[25], G. Desvignes[2], N. Dhanda-Batra[26], C. Dwivedi[27], M. Falxa[6,13], R. D. Ferdman[29], A. Franchini[15,16], J. R. Gair[31], B. Goncharov[32,33], A. Gopakumar[24], E. Graikou[2], J.-M. Grießmeier[13,14], A. Gualandris[28], L. Guillemot[13,14], Y. J. Guo[2], Y. Gupta[34], S. Hisano[35], H. Hu[2], F. Iraci[36,19], D. Izquierdo-Villalba[15,16], J. Jang[2], J. Jawor[2], G. H. Janssen[11,37], A. Jessner[2], B. C. Joshi[34,3], F. Kareem[38,39], R. Karuppusamy[2], E. F. Keane[40], M. J. Keith[41], D. Kharbanda[25], T. Kikunaga[35], N. Kolhe[42], M. Kramer[2,41], M. A. Krishnakumar[2,9], K. Lackeos[2], K. J. Lee[43,64], K. Liu[2], Y. Liu[43,9], A. G. Lyne[41], J. W. McKee[44,45], Y. Maan[34], R. A. Main[2], M. B. Mickaliger[41], I. C. Niţu[41], K. Nobleson[47], A. K. Paladi[48], A. Parthasarathy[2], B. B. P. Perera[49], D. Perrodin[19], A. Petiteau[50,6], N. K. Porayko[15,2,★★], A. Possenti[19], T. Prabu[51], H. Quelquejay Leclere[6,★★], P. Rana[24], A. Samajdar[52], S. A. Sanidas[41], A. Sesana[15,16,17,★★], G. Shaifullah[15,16,19], J. Singha[3], L. Speri[31], R. Spiewak[41], A. Srivastava[25], B. W. Stappers[41], M. Surnis[54], S. C. Susarla[55], A. Susobhanan[56], K. Takahashi[57,58], P. Tarafdar[7], G. Theureau[13,14,59], C. Tiburzi[19], E. van der Wateren[11,37], A. Vecchio[18], V. Venkatraman Krishnan[2], J. P. W. Verbiest[61,9,2], J. Wang[9,62,63], L. Wang[41], Z. Wu[43,9], and P. Auclair[5], E. Barausse[10], C. Caprini[21,22], M. Crisostomi[10], F. Fastidio[28,15], T. Khizriev[30], H. Middleton[18], A. Neronov[6,46], K. Postnov[30,53], A. Roper Pol[21], D. Semikoz[6], C. Smarra[10], D. A. Steer[6], R. J. Truant[15], and S. Valtolina[60]

$$\Omega_{\rm GW}(f) = 3\,\mathcal{A}\,\Omega_*^2\,(\lambda_*\mathcal{H}_*)^2 F_{\rm GW,0}\,S_{\rm turb}(\lambda_* f), \tag{21}$$

where Ω_* is the ratio of the (M)HD turbulent energy density to the radiation one, and $\lambda_*\mathcal{H}_*$ is the ratio of the characteristic length scale of the turbulence, λ_*, to the comoving Hubble horizon $\mathcal{H}_*^{-1}$ at the QCD epoch. The parameter $\mathcal{A} \simeq 1.75 \times 10^{-3}$ is the efficiency of GW production[8], estimated in Roper Pol et al. (2022a). The function $F_{\rm GW,0}$ is the fractional radiation energy density at the epoch of GW generation to its value at the present time. It depends on the temperature scale T_* via the number of degrees of freedom g_*,

$$F_{\rm GW,0} \simeq 8 \times 10^{-5} \left(\frac{10}{g_*}\right)^{1/3}. \tag{22}$$

The spectral shape of the GWB signal, $S_{\rm turb}(f)$, is

$$S_{\rm turb}(\lambda_* f) = \mathcal{B}\,(\lambda_* f)^3\, p_\Pi(\lambda_* f) \times \begin{cases} \ln^2[1 + \mathcal{H}_*\,\delta t_{\rm fin}/(2\pi)], & \text{if } f < 1/\delta t_{\rm fin}, \\ \ln^2[1 + \lambda_*\mathcal{H}_*/(2\pi\lambda_* f)], & \text{if } f \geq 1/\delta t_{\rm fin}, \end{cases} \tag{23}$$

[8] This estimate is conservative since it only considers the decaying stage of turbulence. Numerical simulations find larger values when including a stage of turbulence production (Roper Pol et al. 2020b, 2022b; Kahniashvili et al. 2021).

$$\mathcal{H}_* \simeq 10^{-8}\,\frac{T_*}{100\,\text{MeV}}\left(\frac{g_*}{10}\right)^{\frac{1}{6}}\ \text{Hz},$$

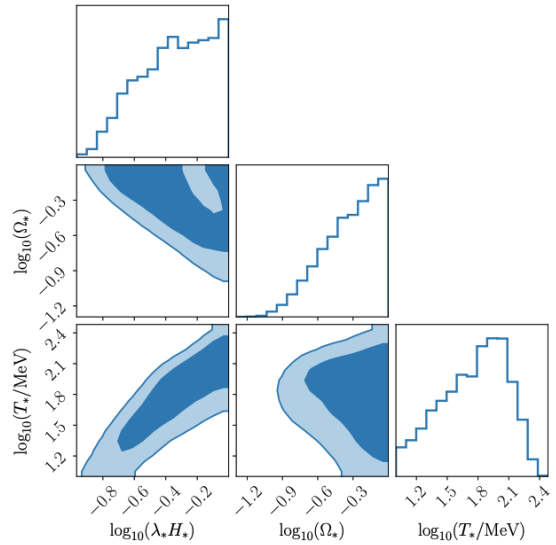
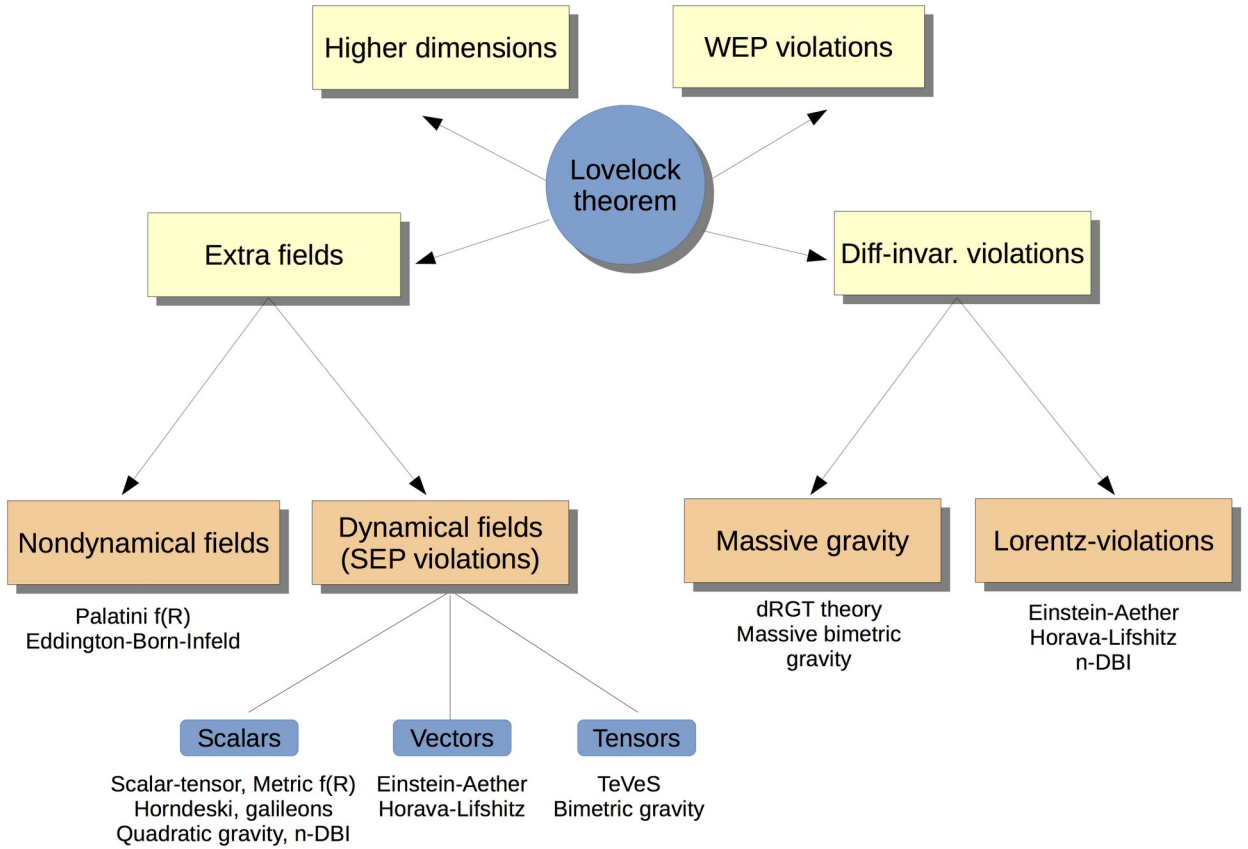
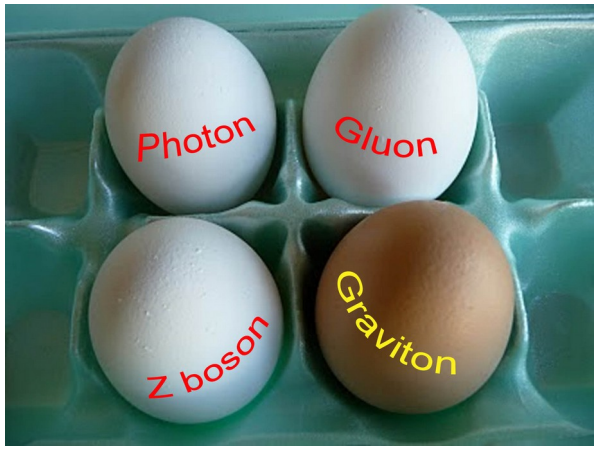


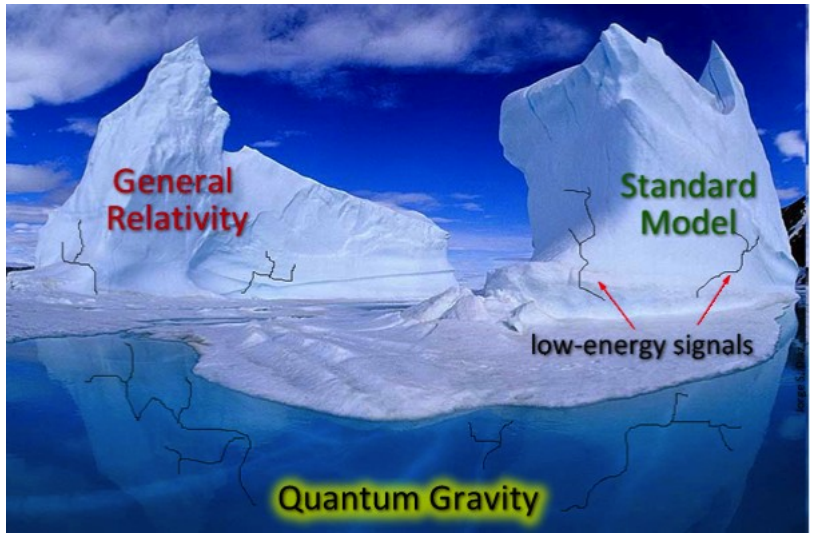
Fig. 18. 2D posteriors for the parameters of the background from turbulence around the QCD energy scale obtained using a free spectrum fit on DR2new data. The 68% and 95% credible regions are displayed.

The GWB produced from vortical (M)HD turbulence is therefore determined by three parameters: the temperature scale T∗, the turbulence strength Ω∗, and the turbulence characteristic length scale λ∗H∗.

NEW PHYSICS – SEARCHING HIDDEN IN THE SEA OF MODELS







Berti et al.

MASSIVE GRAVITY

De Rham 2017

Yukawa

m_g (eV)	λ_g (km)	Eq.	
$\mathbf{7.2 \times 10^{-23}}$	$\mathbf{2.8 \times 10^{12}}$	(49)	A 2σ bound from the precession of Mercury (Talmadge *et al.*, 1988; Will, 1998).
6×10^{-32}	3×10^{21}	(53)	A 1σ bound from weak lensing of a cluster at $z = 1.2$ (Choudhury *et al.*, 2004). Sensitive to the dark matter distribution and cosmological model.
10^{-29}	10^{19}	(52)	From observations of gravitationally bound clusters of 0.5 Mpc (Goldhaber and Nieto, 1974; Hare, 1973). Sensitive to the dark matter distribution.

Dispersion Relation

m_g (eV)	λ_g (km)	Eq.	
$\mathbf{1.2 \times 10^{-22}}$	$\mathbf{1.7 \times 10^{12}}$	(58)	A 90% confidence bound two 30 $M_\odot$ bh-bh merger (GW150916) (Abbott *et al.*, 2016d; Will, 1998).
7.6×10^{-20}	2.6×10^{9}	(65)	From pulsar timing of PSR B1913+16 and PSR B1534+12 (Finn and Sutton, 2002).
10^{-30}	10^{20}	(63)	Observations of power in B–mode polarization in CMB at low ℓ (Dubovsky *et al.*, 2010; Gumrukcuoglu *et al.*, 2012; Raveri *et al.*, 2015).
10^{-26}	10^{16}	(59)	A 10^4 to 10^7 $M_\odot$ merger by eLISA type experiment (Will, 1998).
10^{-24}	10^{14}	(60)	A dual messenger observation of IBWD by eLISA type experiment (Cooray and Seto, 2004; Cutler *et al.*, 2003; Larson and Hiscock, 2000).
10^{-23}	10^{13}	(66)	Pulsar timing array of 100ns accuracy with 10 year observation (Lee *et al.*, 2010).
10^{-20}	10^{10}	(61)	Dual messenger observation of SNe gamma ray burst and gravitational waves (Nishizawa and Nakamura, 2014).

Fifth Force

m_g (eV)	λ_g (km)	Eq.	
10^{-32}	10^{22}	(77)	From earth-moon precession for cubic Galileon theories (Dvali *et al.*, 2003).
10^{-32}	10^{22}	(84)	From precession in full 5D DGP in the Solar System (Gruzinov, 2005; Lue and Starkman, 2003).
10^{-30}	10^{20}	(81)	From earth-moon precession for quartic Galileon theories (dRGT-like) (de Rham, 2014).
10^{-27}	10^{17}	(86)	From PSR B1913+16 pulsar in cubic Galileon theories (DGP) (de Rham *et al.*, 2013d).
10^{-33}	10^{23}	(89)	A prospective 4σ bound from weak lensing on next-gen surveys (Park and Wyman, 2015; Wyman, 2011). Sensitive to alternative DM halo profiles.
10^{-32}	10^{22}	(90)	Observations of altered structure formation from fifth force (Khoury and Wyman, 2009; Park and Wyman, 2015; Wyman, 2011; Zu *et al.*, 2014). Sensitive to the particular theory of massive gravity.

Shankaranarayanan & Johnson

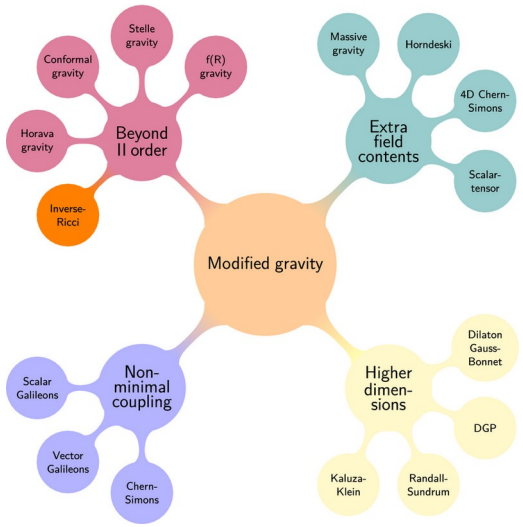
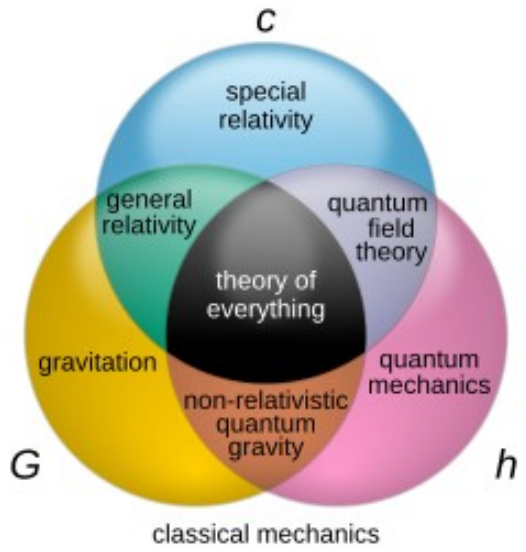
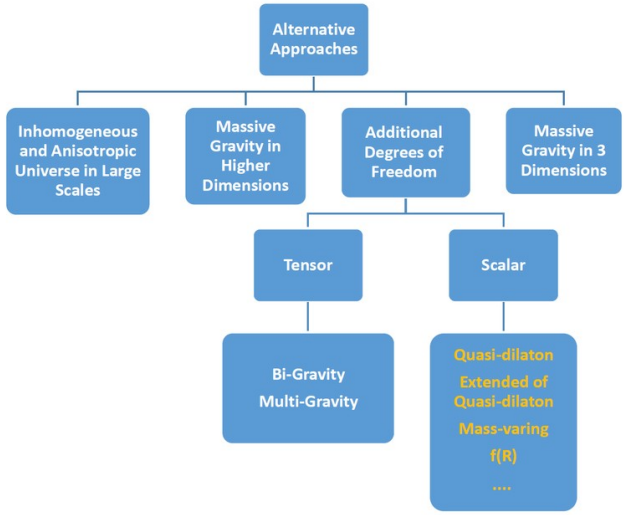


Figure 1. A modified gravity architecture, which gives a representation of some possible ways of modifying GR through breaking the Lovelock's theorem along with some examples. Theories are classified in order to avoid Lovelock's restrictions. In the green region we have theories as de Rham-Gabadadze-Tolley (dRGT) massive gravity theories, and cases with $m_g > 0$. In the red region we have theories with additional field as scalar (Horndeski) which includes cases as Galileon, Quintessence, Brans–Dicke and $f(R)$. Moreover, Vector theories such as Proca and Tensor–Vector–Scalar (TeVes) are presented and, finally, bigravity theories. In the olive region we have an example of theories that are constructed in the break assumption formalism as non-local ones, extra dimensions and Lorentz violating. The main gravitational wave (GW) test of each theory is framed at the top, where the following was considered: GW speed (blue color) and dispersion (red color) refers to constraints from late time GW sources. GW oscillations (yellow color) refer to these oscillations during the propagation introducing a modulation of the GW amplitude. GW damping (green color) refers to probing the damping of GWs using standard sirens from $d_L^{\rm GW}$ (luminosity distance of GWs) and $d_L^{\rm EM}$ (electro-magnetic luminosity distance). Theories that are constrained by these GW observables are enclosed by a dashed square of each GW color, respectively. This figure was designed based on the ideas and discussions behind the construction of theoretically sensible modified gravity theories [10,11,12]. Classification of parameterised post-Newtonian (PPN) tests was based on approximations to obtain the PPN parameters at 1, 2, 4 PN level [13,14,15]. Moreover, a PPN extension for a five-dimensional metric was included [16]. Notice that theories that are in agreement with this PNN test are filled in their respective colors: 2PN level (solid green color), unconstrained (solid red color) and 4PN level/PNN extra-dimensional (olive color).

Escamilla-Rivera and Fabris 2021





Sobhan Kazempour

TESTS OF MASSIVE GRAVITY

Fiertz-Pauli $$S = \int d^4x \left[\frac{1}{2}\partial_\lambda h_{\mu\nu}\partial^\lambda h^{\mu\nu} - \partial_\mu h_{\nu\lambda}\partial^\nu h^{\mu\lambda} + \partial_\mu h^{\mu\nu}\partial_\nu h - \frac{1}{2}\partial_\lambda h \partial^\lambda h + \frac{1}{2}m^2\left(h_{\mu\nu}h^{\mu\nu} - h^2\right)\right]$$

Yukawa-like exponential suppression is expected when the force carrier possesses a nonzero mass. In the case of MG, suppression of the gravitational potential is of the form

$$\Phi \propto \exp\left[-m_g R/R\right]$$

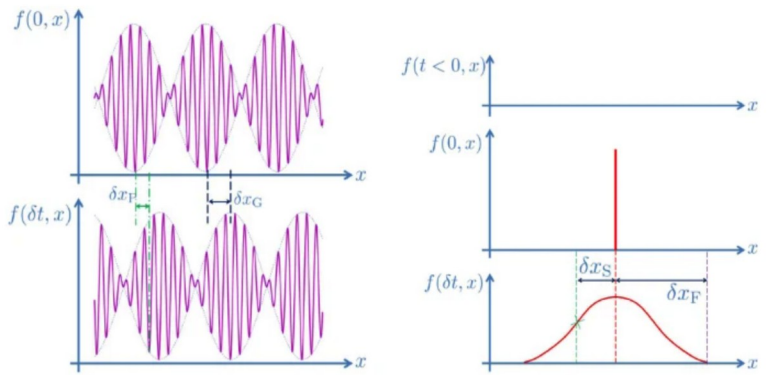
There is also a suppression of GWs at wavelengths larger than the Compton length scale of the graviton $R_g \sim (m_g)^{-1}$

Modification of the dispersion relation gains a massive term, implying a difference between the GW propagation speed c_g and the speed of light c, which can be used to constrain m_g.

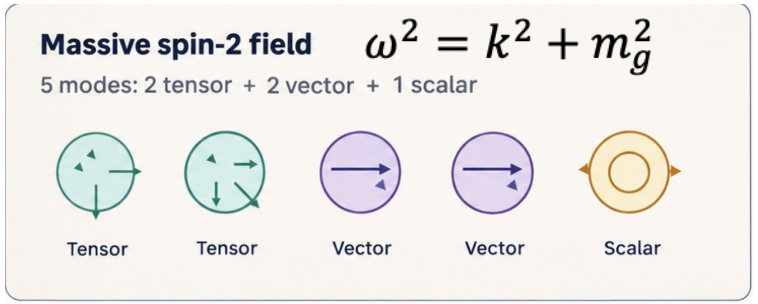
Additional degrees of polarization are characterized by the additional helicity ±1 and 0 modes. Certain theories of MG, such as the minimal theory of MG, only have two tensor modes, as in GR, but a general theory of MG that assumes Lorentz invariance will have five in total, De Rham-Gabadadze-Tolley model

De Rham 2014, Living Review

Figure 5



Difference between phase, group, signal and front velocities. At $t = \delta t$, the phase and group velocities are represented on the left and given respectively by v phase = $\delta x_P/\delta t$ and $v_{\text{group}} = \delta x_G/\delta t$ (in the limit $\delta t \to 0$.) The signal and front velocity represented on the right are given by $v_{\text{signal}} = \delta x_S/\delta t$ (where δx_S is the point where at least half the intensity of the original signal is reached.) The front velocity is given by $v_{\text{front}} = \delta x_F/\delta t$.



PULSAR TIMING ARRAYS DATA AND ADDITIONAL POLARIZATIONS

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The NANOGrav 12.5-year Data Set: Search for Non-Einsteinian Polarization Modes in the Gravitational-wave Background

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The NANOGrav Collaboration

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The NANOGrav 15 yr Data Set: Search for Transverse Polarization Modes in the Gravitational-wave Background

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The NANOGrav Collaboration

The Astrophysical Journal Letters, 923:L22 (18pp), 2021 December 20

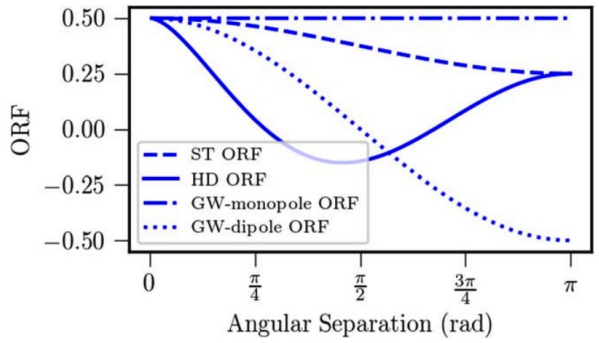


Figure 4. The solid line is the HD correlation curve, the dashed line is the ST ORF, the dashed–dotted line is the GW-monopole ORF, and the dotted line is the GW-dipole ORF. These four ORFs can be hard to distinguish if the uncertainties in the timing residual cross-correlations are sufficiently large.

The analyses in this Letter, as well as those in Bernardo & Ng (2023) and Chen et al. (2023), do not rule out the possibility of ST correlations in our data. However, our analysis also shows no statistical need for an additional stochastic process with ST correlations.

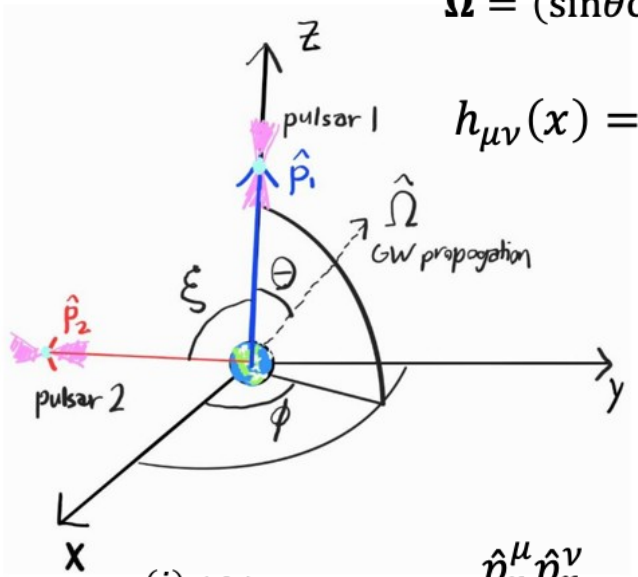
PTA Geometry and the Overlap Reduction Function

How two pulsars encode the angular correlation of a stochastic gravitational-wave background

$\xi_{a\beta}$ — angular separation of pulsars a and b

$_b = (\sin\xi_{ab}, 0, \cos\xi_{ab})$ - unit vectors to

$\hat{\Omega} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$: GW: propagation direction

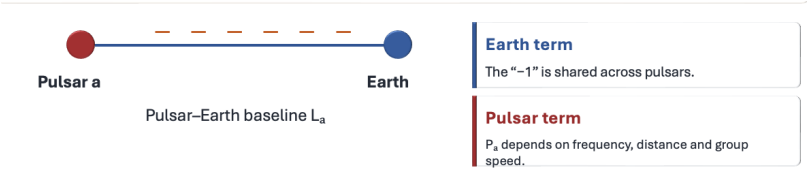


$$h_{\mu\nu}(x) = \frac{1}{2\pi}\int d^4k\, \frac{2\delta\left(\omega^2 - |\mathbf{k}|^2 - m_g^2\right)}{|\mathbf{k}|}\, e^{ikx}\, h_{\mu\nu}(k) \qquad h_{\mu\nu}(k) = \sum_{I\in\{T,V,S\}}\sum_{i\in I} h^{(i)}(k)\, \epsilon_{\mu\nu}^{(i)}$$

Observable: correlated timing residuals

For an isotropic background, the ORF describes the pair's sky-averaged response.

$$\tilde{z}_r(f,\hat{\Omega}) = \sum_{I\in\{T,V,S\}}\sum_{i\in I}\left[e^{-i2\pi f L_r(1+v_g\hat{\Omega}\cdot\hat{\mathbf{p}}_r)} - 1\right] h^{(i)}\left(f, v_g\hat{\Omega}\right) F_r^{(i)}(\hat{\Omega})$$



$$F_r^{(i)}(\hat{\Omega}) = -\frac{\hat{p}_r^\mu \hat{p}_r^\nu}{2\left(1 + v_g\hat{\Omega}\cdot\hat{\mathbf{p}}_r\right)}\epsilon_{\mu\nu}^{(i)} + \hat{p}_r^\mu \epsilon_{0\mu}^{(i)} - \frac{1}{2}\epsilon_{00}^{(i)}\left(1 + v_g\hat{\Omega}\cdot\hat{\mathbf{p}}_r\right)$$

$$v_g \equiv \frac{|\mathbf{k}|}{\omega} = \sqrt{1 - \frac{m_g^2}{\omega^2}}$$

GR tensor limit (distant pulsars): $\tilde{\Gamma}$ → Hellings–Downs correlation

As group speed → 0, the phases lose direction dependence, but not their pair dependence.

Effective ORF: Tensor + Vector + Scalar

The angular correlation depends on polarization weights and finite pulsar phases

$$\tilde{\Gamma}_{ab}^{\mathrm{eff}} = \beta_T \left[\Gamma_{ab}^{T} + \frac{\Omega_V}{\Omega_T} \Gamma_{ab}^{V} + 2 \frac{\Omega_S}{\Omega_T} \Gamma_{ab}^{S} \right]$$

Γ^i – unnormalized angular ORFs of the three sectors

β_T — tensor normalization; Ω_S/Ω_T and Ω_V/Ω_T— relative energy densitie $\mathcal{P}_r(\hat{\Omega}) \equiv e^{-i2\pi f L_r \left(1+v_g \hat{\Omega}\cdot\hat{p}_r\right)}, \qquad \mathcal{P}_0 = 1.$

TENSOR · ±2

Massless limit: reproduces the Hellings–Downs correlation.

VECTOR · ±1

The response scales quadratically with mass near GR.

SCALAR · 0

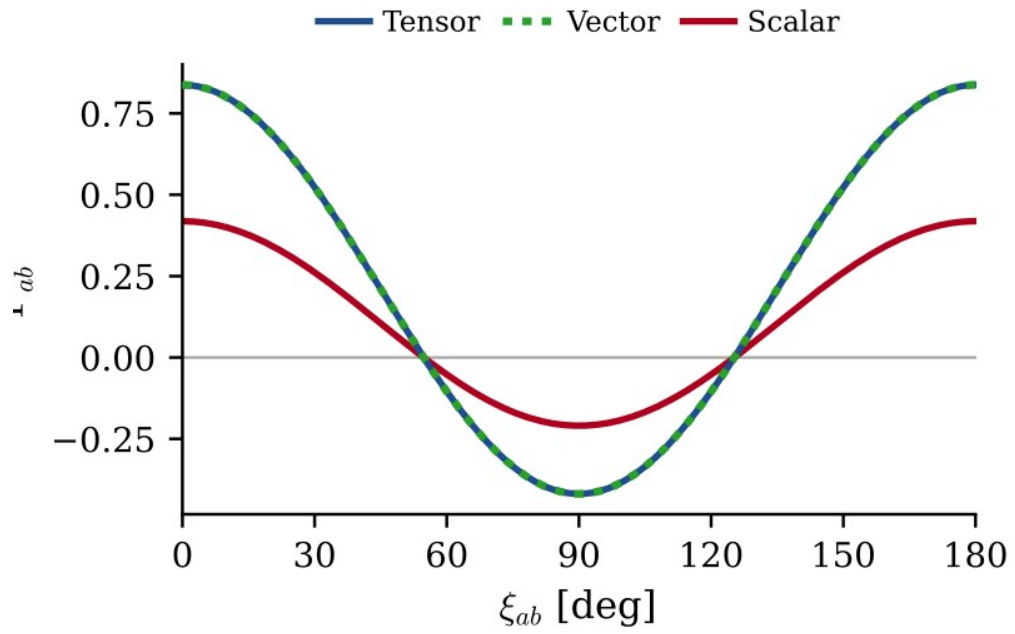
A finite breathing response survives as mass → 0.

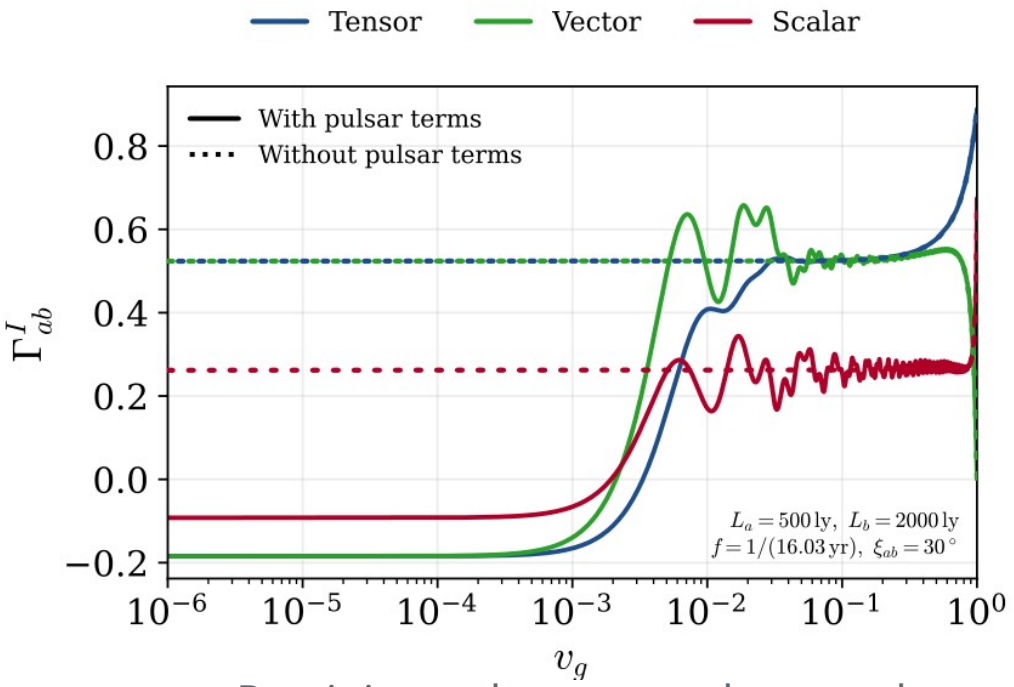
$$\tilde{\Gamma}_{ab}^{I}(\xi_{ab}, f) \equiv \beta_I \Gamma_{ab}^{I} = \beta_I \,\mathrm{Re}\left[\sum_{i \in I} d^2\hat{\Omega}\, (P_a^* - 1)(P_b - 1) F_a^{(i)*} F_b^{(i)} \right]$$

The product has four pieces: Earth–Earth, Earth–Pulsar, Pulsar–Earth, Pulsar–Pulsar.

Massive Limit: Universal Quadrupole

Identical angular form, but the amplitude carries pulsar-pair phase information





All three sectors approach $P_2(\cos \xi_{a\beta})$.

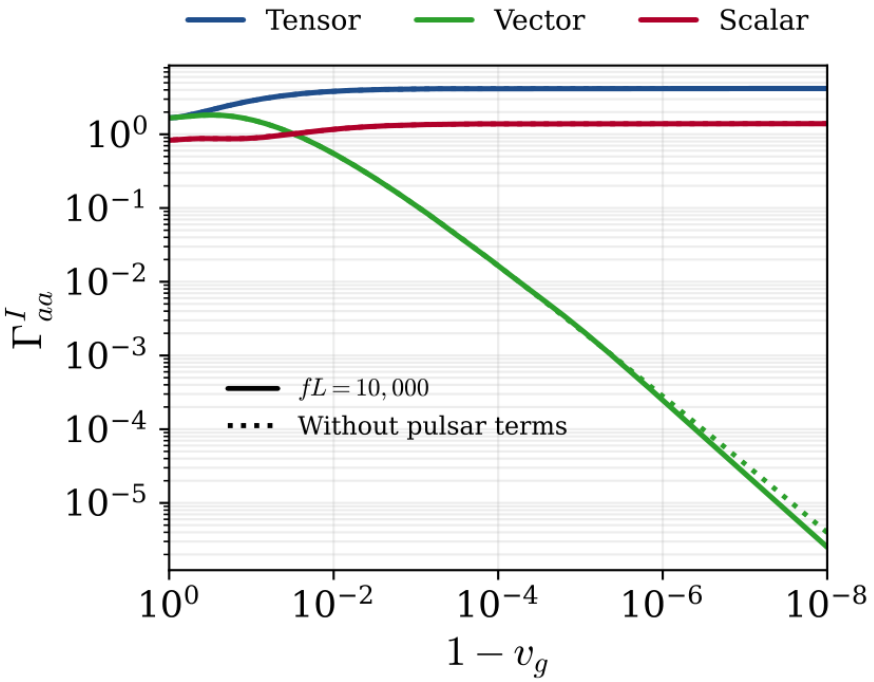
Retaining pulsar terms changes the result.

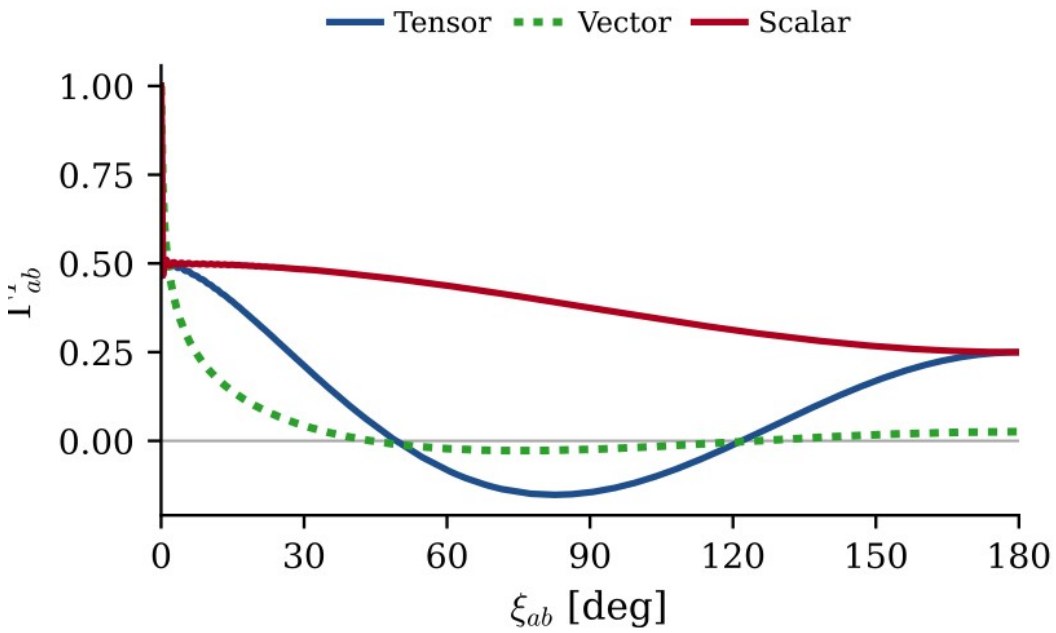
$$\Gamma^T_{ab}(v_g = 0) = \Gamma^V_{ab}(v_g = 0) = \frac{4\pi}{15} R^0_{ab} P_2(\cos \xi_{ab}),$$

$$\Gamma^S_{ab}(v_g = 0) = \frac{2\pi}{15} R^0_{ab} P_2(\cos \xi_{ab}),$$

Massless Limit: the Scalar Does Not Vanish

Why massive spin-2 correlations do not automatically reproduce the GR prediction





The vector autocorrelation decays toward GR.

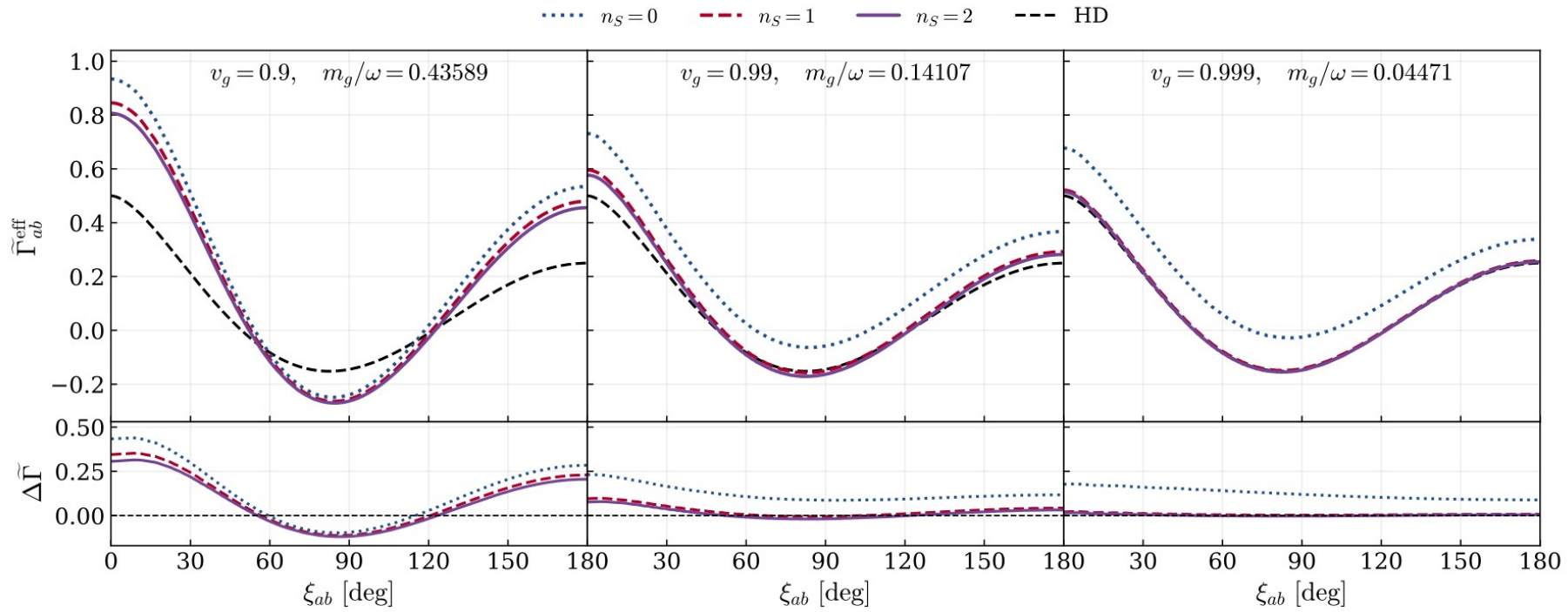
Scalar breathing correlation remains finite.

$$\Gamma^V \sim (m_g/\omega)^2 \qquad \Gamma^S \sim O(1) \qquad \Gamma^T \to \mathrm{HD}$$

Suppressing the Scalar Restores Hellings–Downs

A phenomenological prescription controls continuity as the graviton mass vanishes

$$\frac{\Omega_S}{\Omega_T} = A_S\left(m_g/\omega\right)^{n_S} \; n_s > 0 \;\Rightarrow\; \Gamma \to HD$$



Positive scalar suppression index restores the HD correlation.

From Pulsar-Pair ORFs to the NANOGrav Likelihoo

The same angular response enters the covariance of correlated pulsar timing residuals

$T_{o\beta s}$ = 16.03 yr $f_j = j / T_{o\beta s}$

COMMON-PROCESS COVARIANCE

$$\Phi_{ab}^{GR}(f) = \hat{\Gamma}_{ab}^{GR}(f;\, \xi_{a\beta}, L_a, L_\beta)\Phi_{GR}(f)$$

$$\Phi_{ab}^{MG}(f) = \hat{\Gamma}_{ab}^{eff}(f;\, \xi_{a\beta}, m_g, L_a, L_\beta)\Phi_T(f)$$

Massive gravity: tensor + vector + scalar ORFs

$$\Gamma_{ab}^{\text{eff}} = \beta_T \left[\Gamma_{ab}^{T} + (\Phi_V/\Phi_T)\Gamma_{ab}^{V} + 2(\Phi_S/\Phi_T)\Gamma_{ab}^{S}\right]$$

Earth and pulsar terms retained

TABLE I: Priors used in the Bayesian analysis. The indices a and k label pulsars and backend/receiver systems, respectively. White-noise parameters are fixed to the NANOGrav 15-year values from their public data release [47].

Param.	Description	Prior
E_k	EFAC	Fixed
Q_k	EQUAD	Fixed
J_k	ECORR	Fixed
$A_{\mathrm{RN},a}$	Red-noise amp.	log-U$[-20,-11]$
$\gamma_{\mathrm{RN},a}$	Red-noise spectral index	U$[0,7]$
A_{gw}	C-P amp.	log-U$[-18,-11]$
γ_{gw}	C-P spectral index	U$[0,7]$
m_g/eV	Graviton mass [eV]	log-U$[10^{-25},10^{-23.09}]$
A_S	Scalar-tensor relative amp.	U$[0,1]$
n_S	Scalar suppression index	U$[0,1.7]$

67 pulsars

14 common-process Fourier bins

2,211 distinct pulsar pairs

Massive-Gravity Model: Propagation and Scalar Screening

The parameters are sampled continuously while preserving a smooth GR limit $\omega^2 = k^2 + m_g^2$

$$v_g = \sqrt{1 - \left(\frac{m_g}{\omega}\right)^2}$$

$$\frac{\Phi_S}{\Phi_T} = A_S \left(\frac{m_g}{\omega}\right)^{n_S}$$

$$\frac{\Phi_V}{\Phi_T} = 1$$

PRIORS

m_g	log-uniform 10^{-25} – 8.13×10^{-24} eV
A_S	uniform 0 – 1
n_S	uniform 0 – 1.7

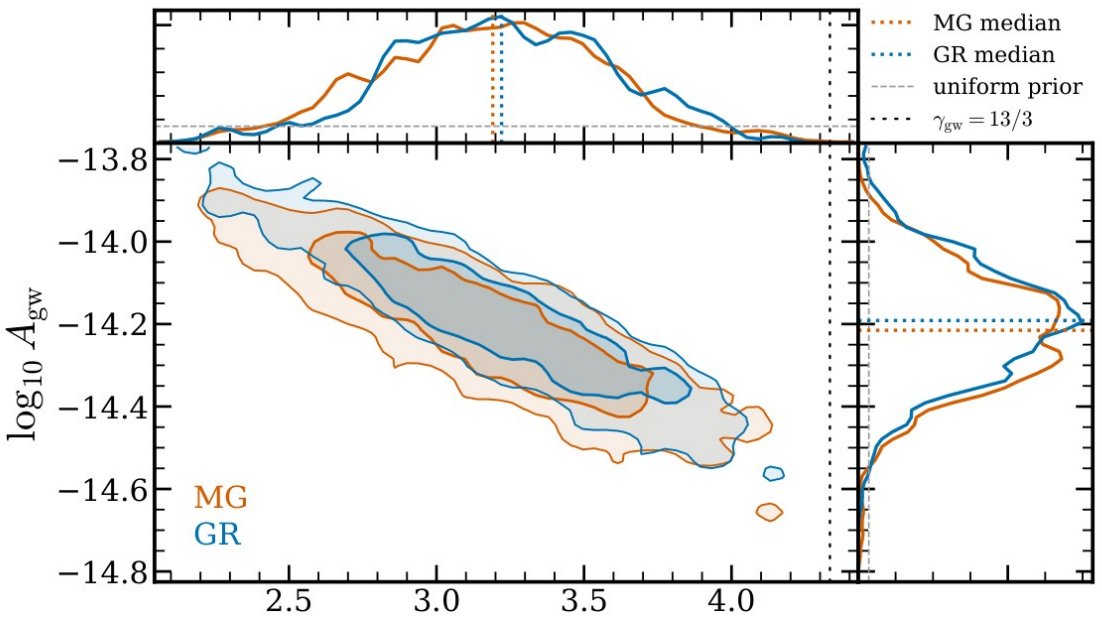
BAYESIAN MODEL

- 67 pulsars; fixed white-noise parameters
- Precomputed ORFs on 3,264 mass values
- Product-space comparison: GR versus MG

$n_S > 0$ suppresses the scalar sector as $m_g/\omega \to 0$, recovering the Hellings–Downs limit.

Common-process posteriors: GR and MG Agree

The amplitude and spectral index are essentially unchanged by the massive-gravity model



68% and 95% posterior contours

MASSIVE GRAVITY

$$A_{gw} = (6.1\ ^{+4.2}_{-2.6}) \times 10^{-15}$$

$$\gamma_{gw} = 3.19\ ^{+0.62}_{-0.64}$$

GENERAL RELATIVITY

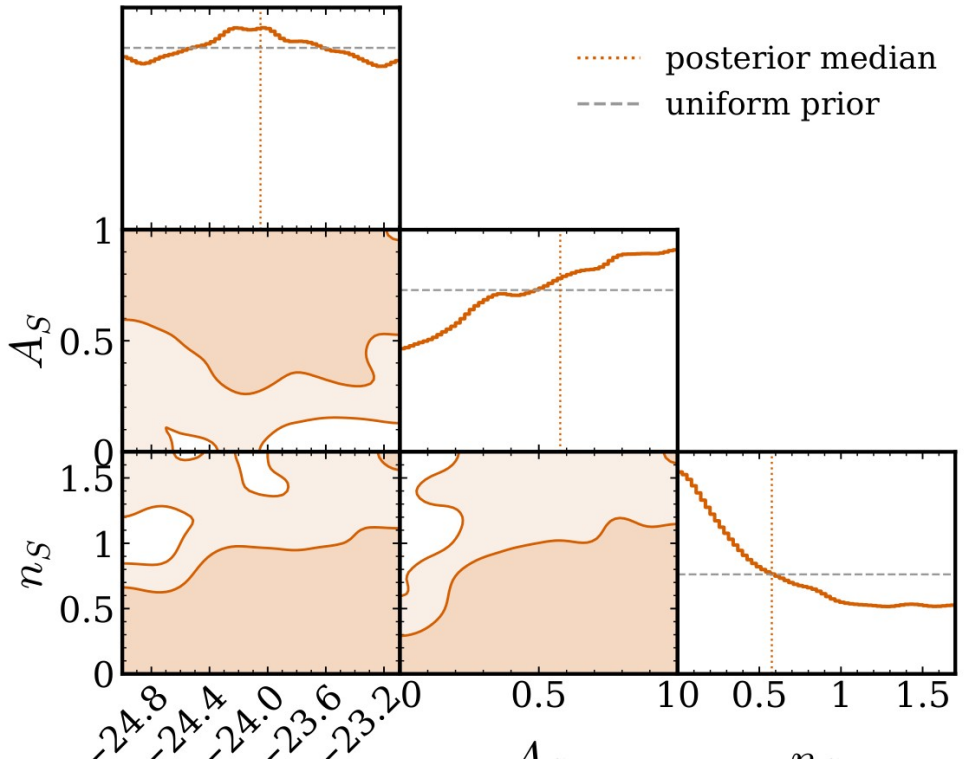
$$A_{gw} = (6.4\ ^{+4.5}_{-2.7}) \times 10^{-15}$$

$$\gamma_{gw} = 3.22\ ^{+0.62}_{-0.61}$$

GR and MG fit the common process equally well.

Graviton Mass: the Posterior Tracks the Prior

Neither the graviton mass nor the scalar parameters are sharply determined



$log_{10}(m_{g/eV})$, A_s, n_s posteriors

BAYES FACTOR $B_{MG/GR}$

1.98 ± 0.28

Weak Bayes-factor preference; not convincing evidence for massive gravity.

GRAVITON-MASS RANGE (90%)

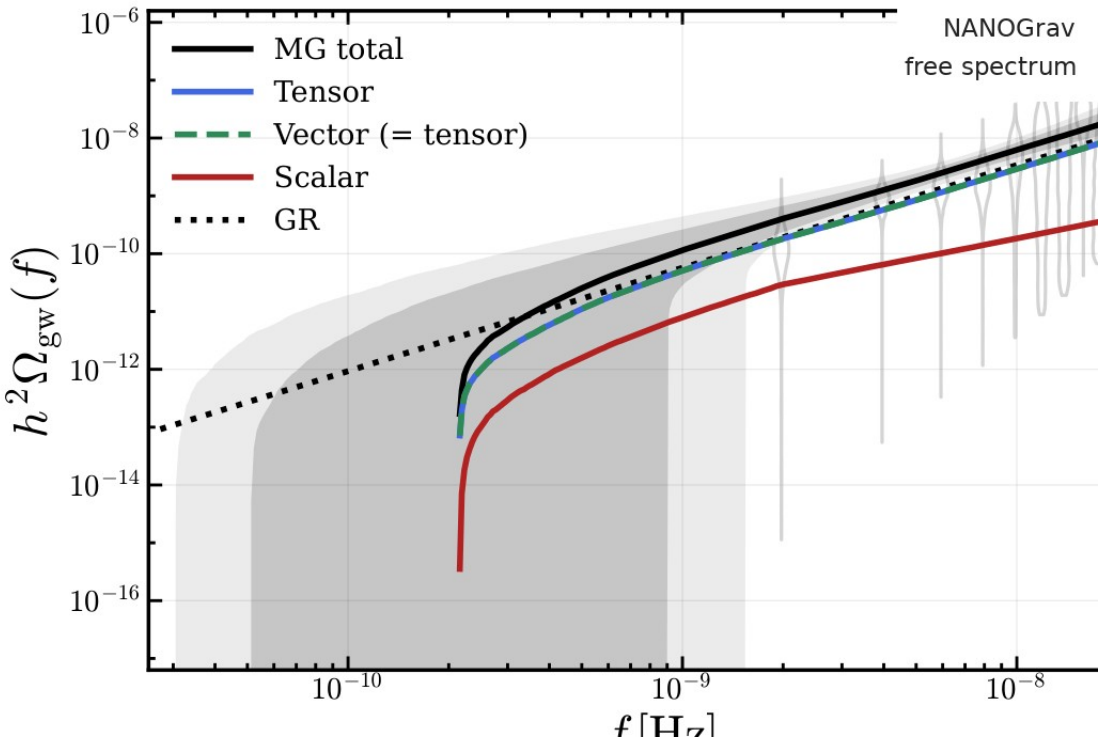
$$\log_{10}(m_{g/eV}) \in [-24.899, -23.193]$$

Prior: [−24.905, −23.186] — nearly identical to the posterior interval.

A_s and n_s also remain broadly distributed.

GW Energy-Density Spectrum and the Low-Frequency Cutoff

MG spectra are compatible with the NANOGrav free-spectrum reconstruction



FMG energy-density spectrum

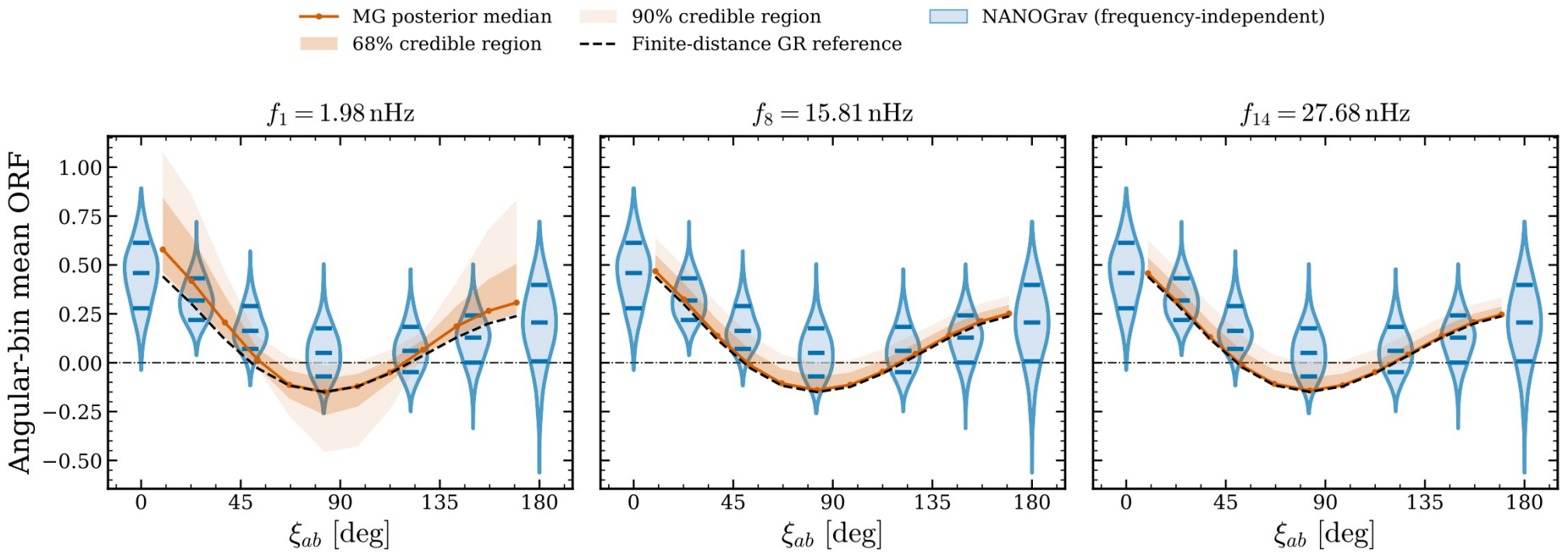
MEDIAN-MASS THRESHOLD

$$f_g = m_g/(2\pi) \simeq 0.22 \text{ nHz}$$

- Lowest sampled Fourier frequency: $f_1 \simeq 1.98$ nHz
- Tensor and vector spectra coincide ($\Omega_V = \Omega_T$).
- The predicted cutoff lies below the observed PTA band.

Frequency-Dependent Correlations Remain GR-Like

Massive-gravity ORFs across three Fourier bins agree with the finite-distance GR reference



f_1 = 1.98 nHz • f_8 = 15.81 nHz • f_{14} = 27.68 nHz

TAKE-HOME: NANOGrav 15-year data do not favor massive gravity; longer baselines can improve sensitivity.

What Can Pulsar Timing Arrays Teach Us?

A new observational window — but not yet a unique identification of the GW source

THE ORIGIN OF NANOHERTZ GWs

- SMBHB populations versus early-universe sources
- Use spectral shapes, anisotropy, and individual signals

THE EARLY UNIVERSE

- Phase transitions, sound waves and primordial (M)HD turbulence
- Constrain source scales, duration and energy budget

TESTS OF GRAVITY

- Angular correlations probe polarization and GW propagation
- Current massive-gravity analysis remains consistent with GR

A stochastic background is evidence for GWs — not yet proof of a particular source.

What Can AI Teach Us?

From costly simulations and noisy data to testable physical predictions

LEARN FROM SIMULATIONS

- Emulate turbulent evolution and the resulting GW spectra

DECODE COMPLEX DATA

- Search for subtle correlations and evaluate noise/systematics

EXPLORE PHYSICAL MODELS

- Speed up parameter inference and compare degenerate scenarios

GUIDE THE NEXT STEP

- Select informative simulations and observations

AI is not evidence by itself:
physical constraints, calibrated uncertainties and validation remain essential.

Take-Home: Physics, Observations & AI

A closed loop connecting fundamental questions to measurable signals

EARLY-UNIVERSE PHYSICS

Phase transitions
Bubbles • sound waves
(M)HD turbulence

→

GW OBSERVABLES

Spectrum and amplitude
Angular correlations
Polarization • anisotropy

→

INFERENCE & AI

Pulsar timing data
Numerical simulations
Uncertainty-aware models

Observations constrain models; simulations interpret signals; AI helps connect both.

The goal is not just to fit the data, but to understand the physics that generated them.

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